BEST 'A' LEVEL BIOLOGY REVISION®

ZIMSEC TOPICAL QUESTIONS AND ANSWERS

Compiled by G. TARUVINGA

BIOLOGY PAPER 2 TOPICAL QUESTIONS & ANSWERS
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ZIMSEC A LEVEL BIOLOGY PAPER 2 TOPICAL QUESTIONS AND ANSWERS

2000 to 2020

Compiled by G. Taruvinga from ZIMSEC past examination papers

Contents

TOPIC 1 CELL STRUCTURE AND FUNCTION

1. **Fig 1.1** shows the structure of the cell membrane.

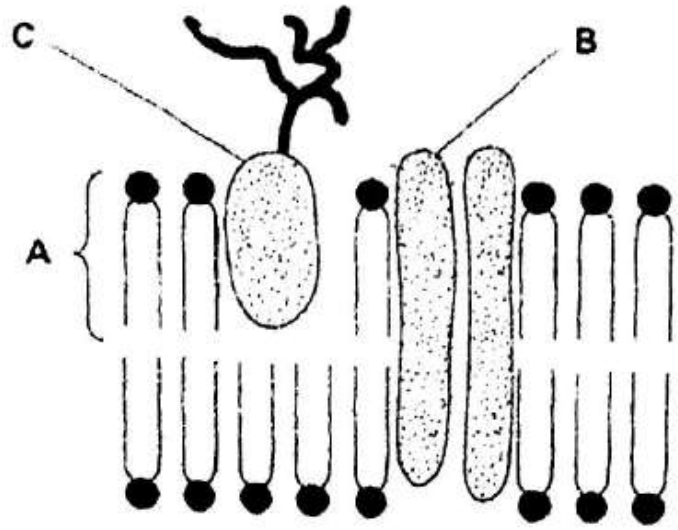


Fig. 1.1

(a) Name the parts labelled **A** and **B**.

A: Phospholipid

B: Channel protein/transmembrane protein [2]

(b) (i) State **one** role of the component attached to **C**.

Cell to cell recognition sites/helps cells to attach to make tissues [1]

(ii) Name **one** organelle that would be found in large numbers in a mucus-secreting cell and describe its role in the production of mucus.

- ***Organelle:*** Secretory vesicles
  Role: transport mucus to the cell surface by exocytosis
- ***Organelle:*** mitochondria
  Role: provide the energy in the form of ATP required for the secretory/exocytosis of mucus.
- ***Organelle:*** endoplasmic reticulum / golgi bodies
  Role: needed to synthesise the proteins found in mucus (e.g. mucin)

ANY ONE [2]

Ref. #1. 6030/2 J2019 NC

2. **(a) Table 1.1** shows some features of cells.
Complete the table by writing 'absent' if feature never occurs, 'present' if it is always present or 'sometimes' if it occurs in some cells but not others. [4]

Table1.1

Feature	Prokaryotic cell	Eukaryotic cell
Plasmid	Present	Absent
Ribosomes	Present	Present
Flagellum	Sometimes	Sometimes
Cell wall	Present	Sometimes

(b) Explain the difference in resolution between the electron microscope and the light microscope.

Wavelength of electron is smaller than that of visible light; so electron microscope resolution is higher than that of light microscope. [2]

(c) Table 1.2 shows biological structures and their sizes.

Table 1.2

Structure	Size
Virus	100 nm
DNA molecule	2 nm
Bacterium	1µm
Plant cell	100µm
Actin molecule	3.5 nm

State all the structures in **Table 1.2** that can be resolved by a:

(i) Light microscope

Plant cell and bacterium [1]

Ref. #1ScA. 9190/2 J2018

3. (a) Explain the effects of temperature on enzyme action.

- enzymes have an active site;
- that fits the substrate precisely;
- changes in temperature can lead to a shape / conformational change in the protein;
- leading to a change in the shape of the active site;
- may interfere with the binding of the substrate with the active site;
- increase in temperature can increase molecular motion leading to disruption of intermolecular interactions;
- more kinetic energy / faster movement of molecules means more collisions between enzyme / active site and substrate;
- increases chance of enzyme substrate collisions so enzyme activity increases;
- optimal temperature;
- optimum temperature is temperature at which rate of enzyme-catalyzed reaction is fastest;
- at high temperatures enzymes are denatured and stop working;

- denatured means change of structure in enzyme / protein resulting in loss of its biological properties / no longer can carry out its function;
- too much kinetic energy / vibrations breaks bonds that give enzyme specific shape; max [8]

(b) Outline the effects of competitive and non- competitive inhibitors on enzyme activity.

Non-competitive inhibitor

- inhibitor binds (to the enzyme)
- away from the active site / at allosteric site;
- shape / (intramolecular) bonding / conformation of the protein / enzyme is altered;
- shape / properties of active site altered;
- substrate no longer fits the active site / no enzyme-substrate / ES complex formed;
- no enzyme activity / works more slowly (until the inhibitor dissociates);
- eg CN inhibition of cytochrome oxidase by binding to SH groups / other valid example;
- **allosteric inhibition** is a form of non-competitive inhibition;
- most allosteric enzymes have multiple allosteric sites;
- metabolites can act as allosteric inhibitors of enzymes earlier in a metabolic pathway to regulate metabolism;
- binding (of end product) to an allosteric site changes shape of enzyme;

Competitive inhibitor

- a molecule structurally similar to the substrate binds to the active site;
- preventing substrate binding;
- Energy Key: original reaction with enzyme Progress of reaction
- eg inhibition of butanedioic acid (succinate) dehydrogenase by propanedioic acid (malonate) in the Krebs cycle / other valid example;
- competitive inhibition is reversible; max [8]

Ref. #10ScB. 9190/2 J2018

4. (a) Describe the preparation of a sample of tissue for examination with the electron microscope.

- **Fixation;** arrests all cellular activities/preserve morphological and chemical composition.
- **Dehydration;** to prevent decay by removing all water

- **Clearing;** makes material transparent/removes alcohol.
- **Infiltration;** support specimen/for sections of sufficient thichness to be cut.
- **Embedment;** supporting the specimen when being sectioned.
- **Sectioning;** to be able to view the specimen as it will be thin.
- **Staining;** to obtain contrast between different structures.
- **Mounting** (15 max 8) [8]

4.(b) Distinguish between the lysosome and the ribosome in terms of structure and function.

- **Structure**

Lysosome	**Ribosome**
Size/diameter 0.1μm to 0.5μm/Larger than ribosomes	Diameter 20nm/sdmaller than lysosomes.
Single membrane/membrane-bound	No membran/not membrane-bound
Contains digestive enzymes	Contains proteins and rRNA
No sub-units	2 sub-units; larger and smaller
One type	2 types; 70S and 80S.
Found in cytoplasm.	Found on RER, inside mitochondria, inside chloroplasts and free in cytoplasm

- **Function**

Lysosome	**Ribosome**
Digest a wide variety of substances/digest polymers and degrade old cell materilas.	Assist in protein synthesis/Synthesise proteins from mRNA.
Assist in exocytosis to release its enzymes out of cell/autolysis	Ribosomes cannot assist in this manner
Can digest cytoplasmic organelles and other membranes/autophagy	Ribosomes cannot do this
Have rejuvinating role in biology of organisms	Vital for manufacture of substances needed by cell to survive

[8]

Ref. #10ScB. 9190/2 J2017

5. Fig 1.1 shows a model of model of the cell membrane.

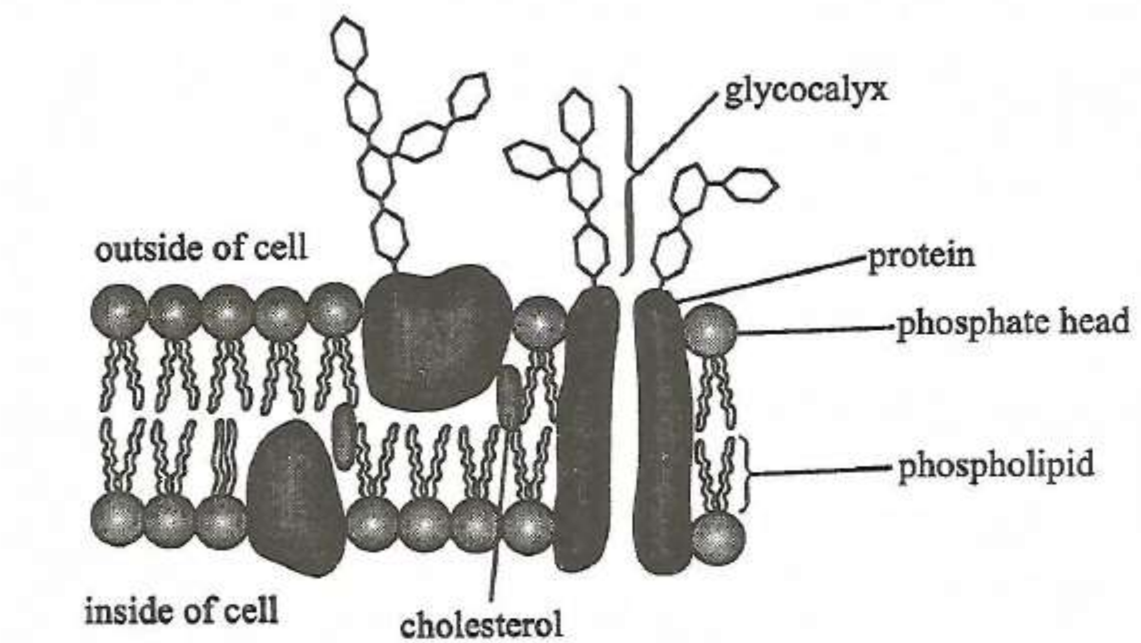


Fig. 1.1

(a) (i) Outline how the membrane structure helps to control movement of water soluble substances across it.

- Hydrophobic tails/fatty acids stop the free movement
- Cholesterol help tom prevent unnecessary leakage of substances;
- Hydrophilic pores/channels allow passage of polar molecules; [2]

(ii) Suggest a reason why the glycocalyx is only found on the outside of the cell surface membrane.

- Glycocalyx serves for cell recognition/ so needs to be outside of membrane where cells are able to detect it;
- So allowing grouping of tissues/cell adhesion; antigens/receptor sites for hormones. [1]

(iii) Explain why glucose molecules cannot pass across the cell membrane by simple diffusion.

- Glucose is a large molecule;
- Polar molecules/water soluble/not lipid soluble [2]

(b) (i) In animal cells, folds in the cell membrane may form microvilli.
Explain the importance of microvilli in cell membranes.

- Increase surface area;
- for passage of substances/absorption/secretion/anchorage during fertilisation by sperm. [2]

(ii) Suggest a reason why microvilli are only found in animal cells but not plant cells.

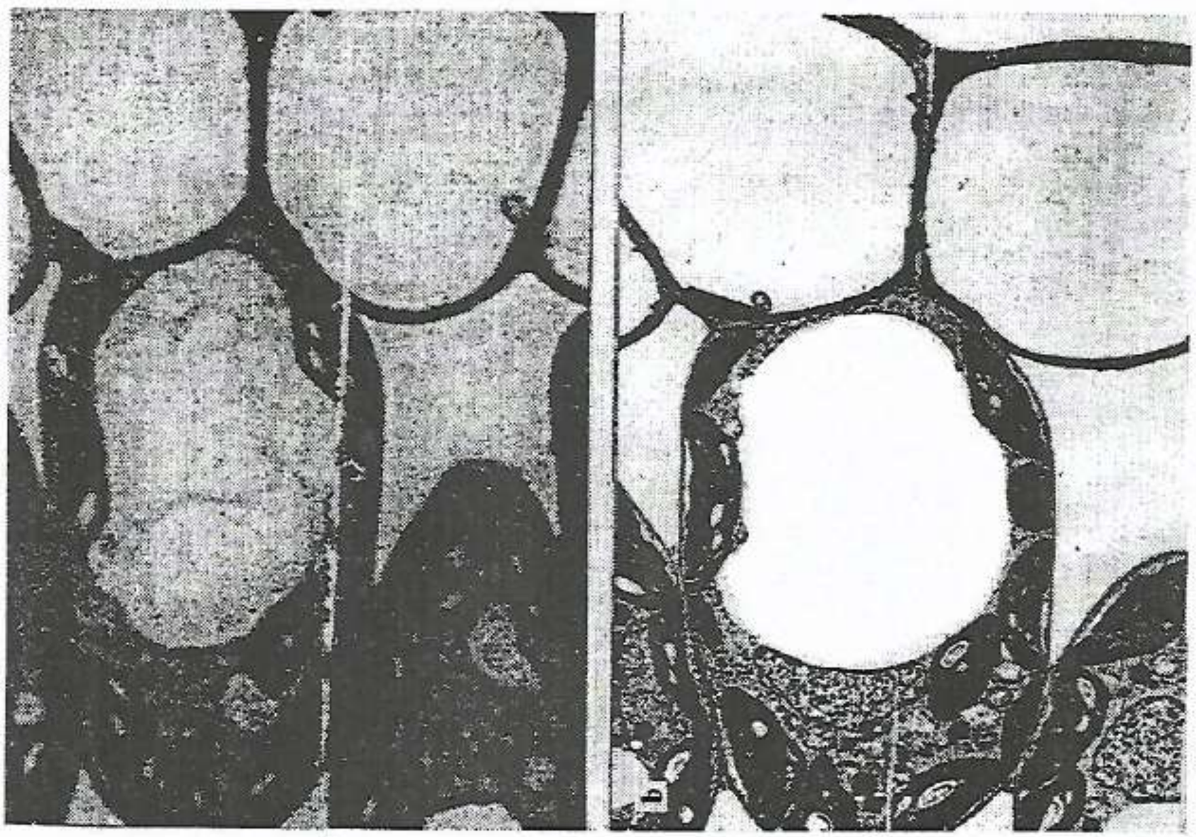
The rigid cellulose cell walls; prevents the membrane folding into microvilli in plant cells/animal cells do not have cell walls; which prevent membrane from folding. [2]

Ref. #1 ScA 9190/2 N2016

6. (a) Define the term

1. *resolution*, Ability to distinguish between two close/adjacent points/objects
2. *magnification*, How many times larger an image is than the specimen [2]

(b) Fig 1.1 is a photomicrograph of plant cells at the same magnification of x400. Photomicrograph **A** was observed under the light microscope and photomicrograph **B** under the electron microscope.



A B

Fig.1.1

With reference to **Fig 1.1**, explain why the actual magnification of an image is less important than the resolution of the image by a microscope.

Does not increase amount of detail/cause blurring of image/does not show clarity between very close points. [2]

(c) **Table1.1** shows procedures, substances and instruments used when preparing material for light and electron microscopy.

Complete **Table 1.1**.

	Substances/instruments for	
Procedure	Light microscope	Electron microscope
Embedding	Wax	Resin/plastic
Fixation	Ethanol/alcohol/acetic acid	Glutaraldehyde/osmic acid/ osmium oxide/formaldehyde
Sectioning	Scalpel blade/Razor blade/ microtone/metal knife	Ultra microtone/glasstone/ diamond blade/
Staining	Coloured dyes/ Iodine/methyl blue / acetic orcein	Heavy metals/ salts of heavy metals (named salt)

[3]

Ref. #1ScA. 9190/2 N2015

7. Fig 1.1 is a model of a section of a cell surface membrane.

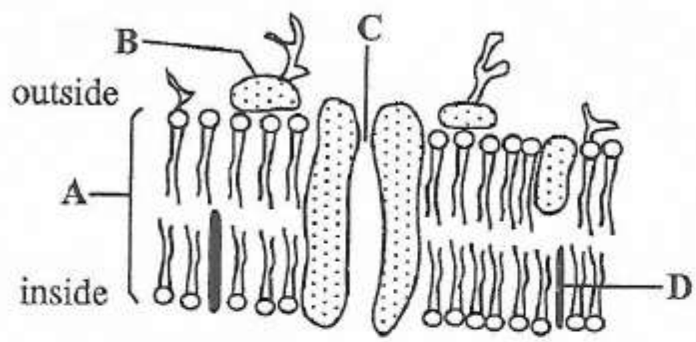


Fig. 1.1

(a) (i) Name the model of the membrane structure shown.

Fluid mosaic model [1]

(ii) Identify the structures labelled **A, B** and **C**.

A Phospholipid bilayer

B Glycoprotein

C ion pore/protein channel. ACCEPT channel/pore [3]

(b) State any **two** functions of the structure labelled **D**.

1. (Cholesterol) reduces lateral movement of other molecules in the membrane.
2. (Cholesterol) adds strength to the membranes/stabilizes phospholipids/disturbs close packing of phospholipids.
3. (Cholesterol) prevents leakage of water and dissolved ions from the cell
4. (Cholesterol) regulates the fluidity of the membrane/Keeps phospholipids fluid.

ANY TWO [2]

(c) Outline four reasons why transport of substances across membranes is vital to a cell.

1. To excrete waste substances;
2. To secrete useful substances;
3. To obtain nutrients;
4. To generate ionic gradients for action potentials/muscle contraction;
5. To maintain suitable pH/ionic concentration. 5 max [4]

(d) Suggest one property that influences the ability of a substance to pass through a cell surface membrane.

Lipid solubility/water solubility/ polar or non-polar/size;

Specific sites; ACCEPT specific examples. [1]

Ref. #1ScA. 9190/2 N2014

8. **Fig. 1** is a diagram of a bacterium as seen under an electron microscope.

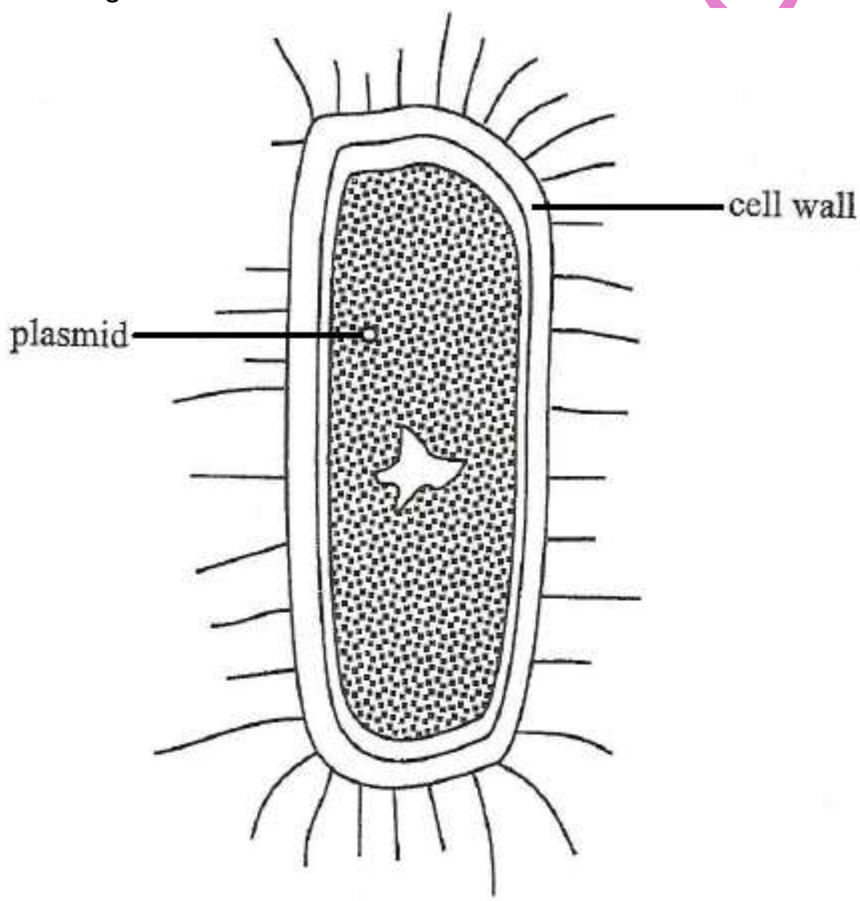


Fig. 1

(a) The bacterium contains DNA, as do eukaryotic cells.

(i) State two other ways in which the structure of the cell in **Fig. 1** is similar to a typical animal cell.

1. Presence of cell (surface) membrane;
2. Presence of ribosomes;

3. Presence of cytoplasm. [2]

(ii) Describe how bacterial DNA differs from that found in an animal cell.
Bacterial DNA consists of circular strands while eukaryotic DNA is linear;
Bacterial DNA has no chromosomes while eukaryotic cells have chromosomes;
Bacterial DNA lies free in cytoplasm while that of eukaryotes is enclosed in the nuclear envelope. [3]

(b) Some bacteria similar to that shown in **Fig. 1** can cause disease. Antibiotics are often given to patients who are suffering from disease caused by bacteria. Examples of the mode of action of two antibiotics are given below:

Antibiotic 1: binds to the enzyme RNA polymerase in bacteria preventing transcription.

Antibiotic 2: prevents the formation of peptide cross links between peptidoglycan chains in the cell wall.

(i) Explain why the prevention of transcription leads to the death of bacteria.

- This prevents formation of further copies of bacterial RNA;
- Because the proteins/enzymes for formation of new bacteria have been prevented from being produced, hence no metabolism

[2]

(ii) Suggest how the action of **antibiotic 2** on cell walls leads to the death of bacteria.

- By allowing increased entrance of harmful/toxic substances into the cells;
- Results in loss of support of protoplasm/ bacteria cannot resist turgor pressure;
- Allowing increased water to enter cells by osmosis; which results in lysis/bursting of cells/mechanical damage. [3]

Ref. #1ScA. 9190/2 N2013

9. Fig 2.1 shows a transverse section of a root module of a legume. **Fig 2.2** is a drawing of a cell from the centre of a nodule from an electron micrograph.

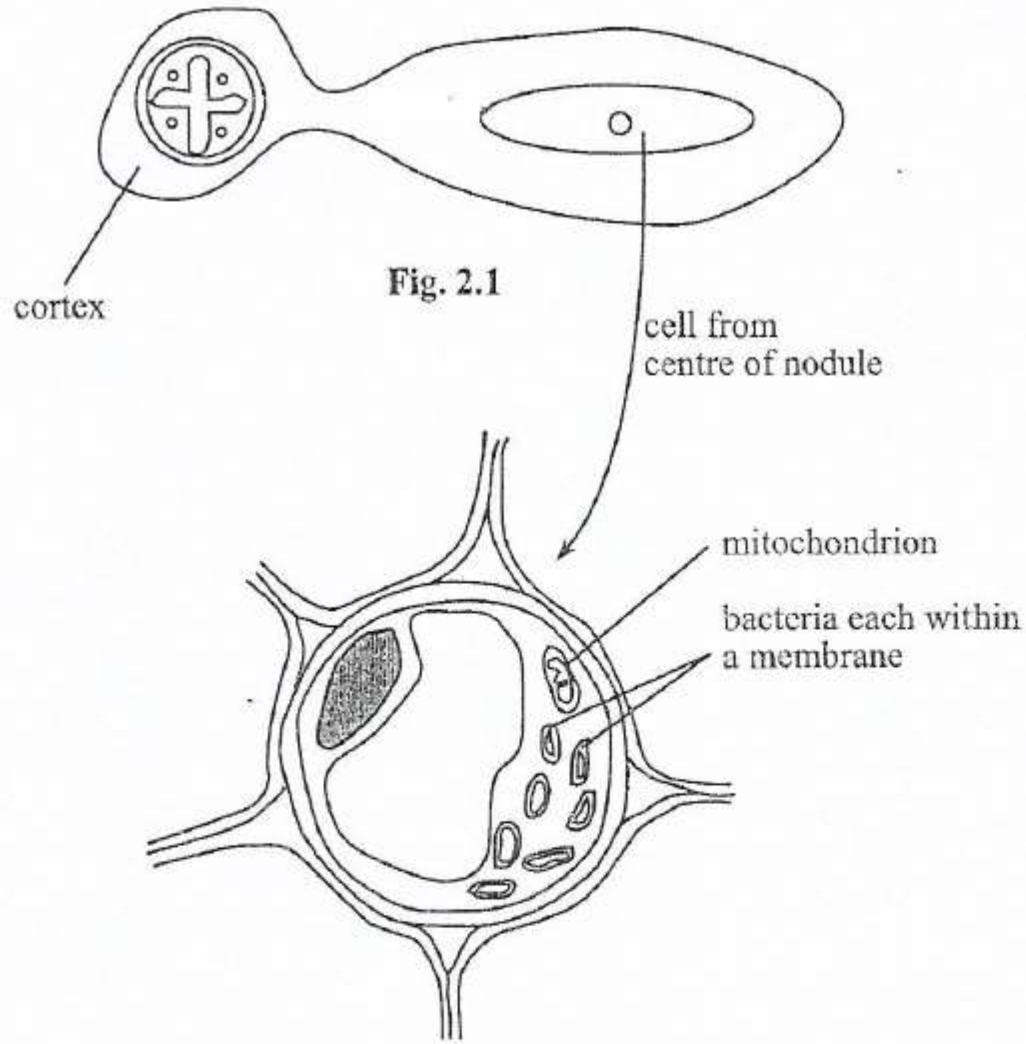


Fig. 2.1

Fig. 2.2

(a) Name type of cells that are present in a cortex of the root that are not present in bacterial cells.

Parenchyma [1]

(b) State **two** advantages of studying cell structure with an electron microscope rather than with a light microscope.

1 High resolution with electron microscope so more detail observed than with light microscope.

2 higher magnification achieved with electron microscope than with light microscope. [2]

(c) Describe the role of *Rhizobium* in the root nodule.

In symbiotic relationship with plant; converts atmospheric nitrogen to ammonia. ACCEPT nitrogen fixation. [2]

Ref. #2ScA. 9190/2 N2011

10. (a) Describe how membrane structure is related to the transport of materials across a membrane.

- Fluid mosaic model structure
- Phospholipid bilayer
- Phospholipids consist of fatty acid chains and a phosphate head
- Fatty acid chains are hydrophobic, while phosphate heads are hydrophilic (orientated towards cytoplasm)
- Allows passage of hydrophobic/nonpolar substances across the bilayer
- Can allow small hydrophilic/polar molecules such as water to pass through/large polar molecules cannot cross
- Cholesterol within the membrane influences the fluidity of the membrane
- (Intrinsic and extrinsic) proteins occur throughout membrane
- Some are receptors for specific molecules
- Glycocalyx is comprised of carbohydrate attached to proteins or lipids
- Proteins may be carriers for specific molecules
- (Gated) channel proteins allow passage of polar/water-soluble substances
- Some cells/organelles contain more protein to act as carriers/receptors depending on their role
- Exocytosis involves transport out of the cell, endocytosis involves transport into the cell;
- In exocytosis, a substance is packaged within a vesicle;
- Vesicle moves to and fuses with the plasma membrane;
- Endocytosis may be phagocytosis or pinocytosis;
- Phagocytosis involves the transport of solid material (into the cell);
- Pinocytosis involves the transport of fluid (into the cell);
- Cell surface membrane invaginates/infolds around the substance to form a vesicle; max [6]

Ref. #10(a)ScB. 9190/2 N2011

11. **Fig 1.1** shows a drawing of an animal cell as seen using an electron microscope.

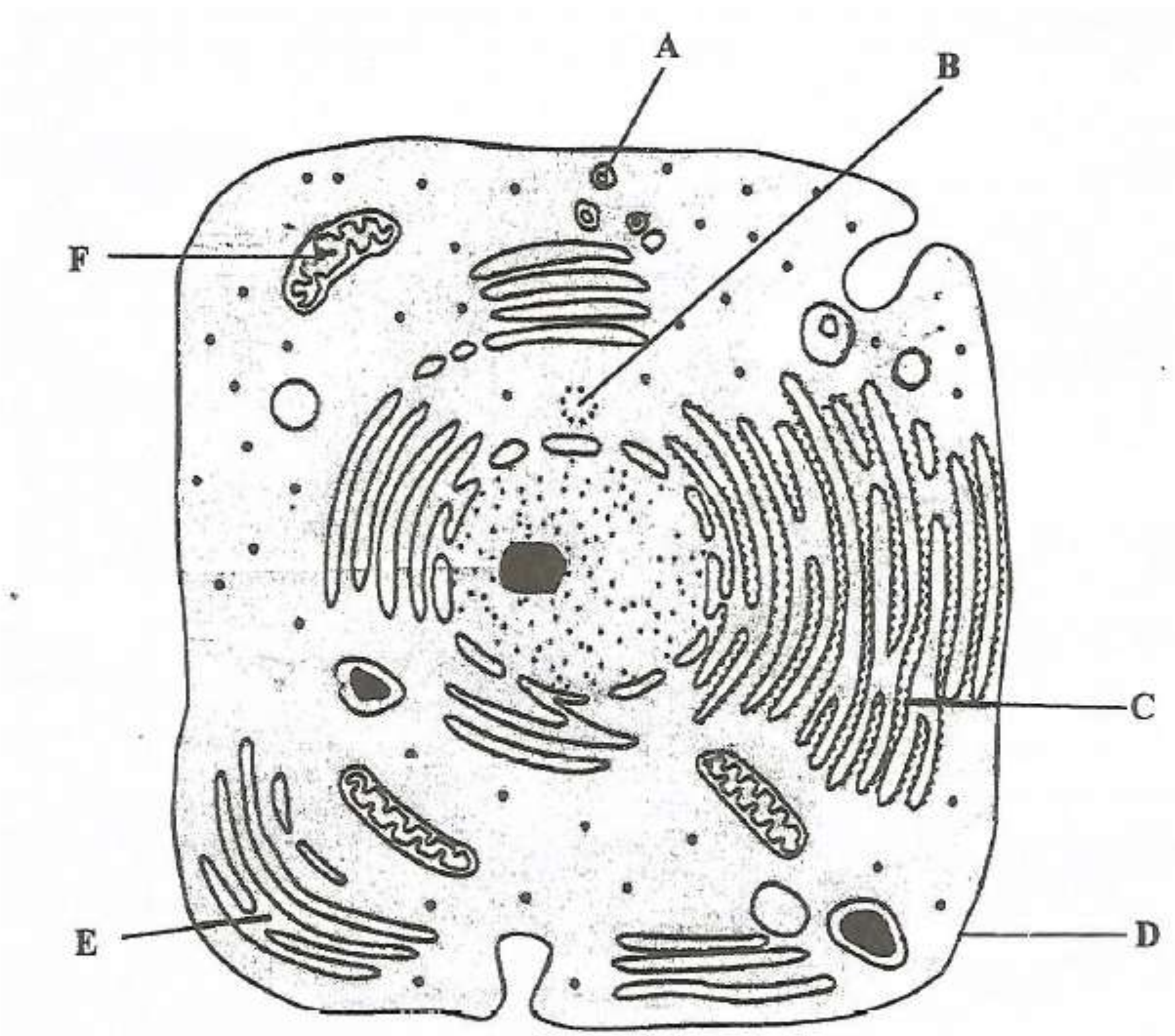


Fig. 1.1

(a) Name the organelles **A, B** and **C.**

A. Secretory vesicle

B. Centriole

C. Rough endoplasmic recticulum/RER [3]

(b) Identify the letter representing the organelle that

(i) doubles just before mitosis,

B [1]

(ii) is responsible for lipid synthesis.

E [1]

(c) Complete the table by writing **yes** or **no** in front of every statement.

statement	yes/no
the structure labelled F is present in both animal and plant cells	Yes
the structure labelled D is the outermost layer in both animal and plant cells	No
the structure labelled C is involved in transcription.	No

[3]

Ref. #1ScA. 9190/2 J2016

12. There are two types of microscopes commonly used in the study of cells. One is the light microscope and the other is the electron microscope. The electron microscope has a higher resolving power than the light microscope.

(a) (i) Define the term resolving power with reference to a microscope.

____

____ [1]

(ii) Explain why an electron microscope has a greater resolving power than a light microscope.

____

____

____ [2]

(iii) Explain why the actual magnitude of an image is less important than the resolution of the image by a microscope.

____

____

____ [2]

(b) Describe how you would measure the diameter of a pollen grain using a light microscope.

____

____

____

____ [3]

Ref. #1ScA. 9190/2 N2005 Silver jubilee

13. **(a)** Contrast the structure of a prokaryotic cell with that of a eukaryotic cell. [8]

(b) Briefly describe the structure of a bacterial cell membrane. [4]

Ref. #10ScB. 9190/2 N2004

14. **Fig 1.1** is a representation of transport methods across a cell membrane.

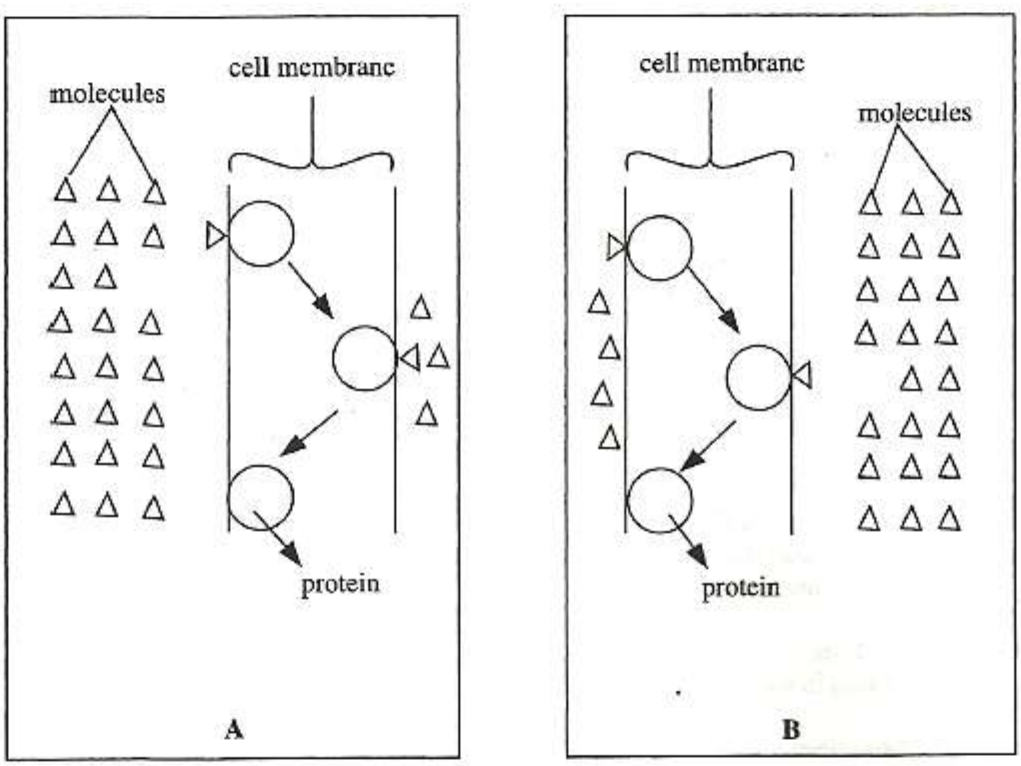


Fig. 1.1

(a) (i) Name the transport method used in A and B.

A. ______________________________

B. ______________________________ [2]

(ii) Describe the transport method taking place in A.

____

____

____

____

______________________________ [3]

(b) Suggest two major differences in transport between A and B not shown in the diagrams.

____

____

____

______________________________ [2]

Ref. #1ScA. 9190/2 J2006

15. (a) Outline the advantages of light microscopy in examining biological specimens. [6]

(b) Relate the structure of a mitochondrion to its function. [6]

Ref. #10ScB. 9190/2 N2008

16. **Fig 1.1** is a diagrammatic model of a cell surface membrane.

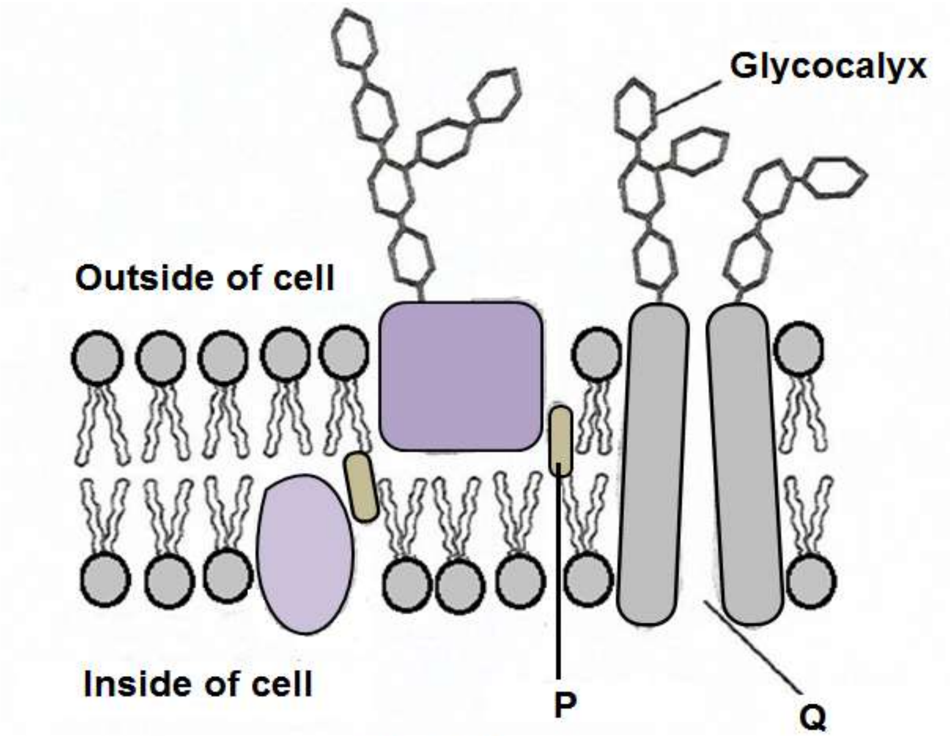


Fig. 1.1

(a) Name the structures labelled **P** and **Q**.

P Cholesterol

Q Hydrophilic channel/Protein pore/protein channel [2]

(b) State the function of the glycocalyx.

- For cell recognition; letting similar cells group together to from tissue.
- Recognition sites e.g. for neurotransmitters and hormones/chemical signals act as antigens .
- ACCEPT Receptor sites/cell adhesion. [1]

(c) Give **two** reasons why the model in **Fig 1.1** is known as fluid mosaic model.

- The phospholipid bilayer is fluid;
- The proteins molecules float about/are scattered resembling a mosaic. [2]

(d) State three features of a substance which determined its ability to pass through a cell membrane.

- Size;
- Fat or water solubility/polar or non-polar/charge;
- Specific sites – which allow the carrier proteins in the cell membrane to recognise the substance

REJECT concentration. [3]

Ref. #1ScA 9190/2 N2009

TOPIC 2 BIOLOGICAL MOLECULES AND WATER

1. **Fig 1.1** shows protein tertiary structure.

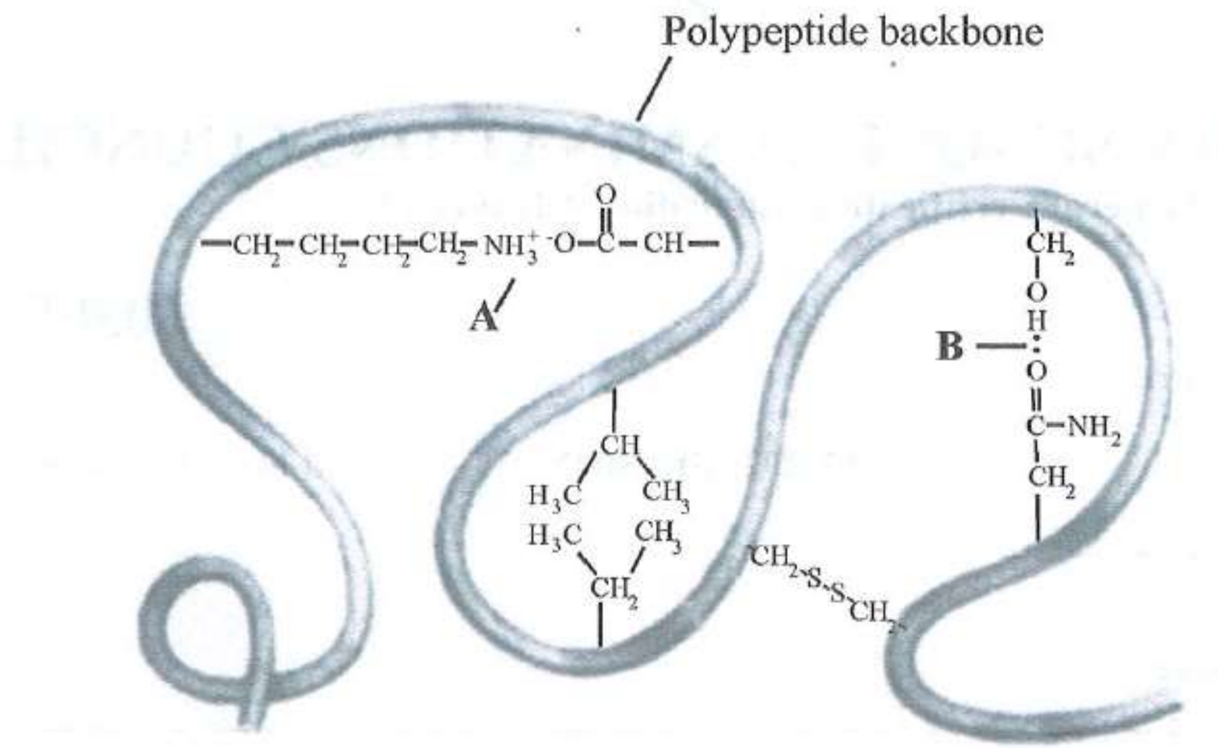


Fig. 1.1

(a) Identify the bonds labelled

A - Ionic bond

B - Hydrogen bond [2]

(b) Name the strongest bond among the bonds shown in **Fig 1.1**.

Disulphide

(c) Describe the difference between globular proteins and fibrous proteins. [4]

Globular	Fibrous
1. Irregular amino acid sequences	Repetitive amino acid sequences
2. Spherical/rounded/ball shaped	Long strands/long and narrow strands
3. Forma a compact structure	Forms strands/lie side by side to form fibres.
4. Have hydrophilic groups on the outside	Have hydrophobic groups on the outside
5. (Mostly) Soluble in water	(Mostly) Insoluble in water
6. Less chemically stable	Chemically stable.
7. Not tough/less tensile strength	Physically tough/high tensile strength
8. Tertiary structure important/ have tertiary or quaternary structures	Secondary structure important/ fibrous have little or no tertiary structure
9. Metabolic role e.g. Enzymes/plasma proteins/Antibodies	Structural role/support functions e.g. collagen in bone and cartilage/Keratin in nails and hair.

Ref. #1. 6030/2 J2020 NC

2. Explain the importance of the following properties of water to living organisms:

(a) A large latent heat of vaporization.

High amounts of heat energy are needed to overcome hydrogen bonds to vaporize a small quantity of water.

Makes water very efficient in cooling process, e.g. transpiration in plants and sweating in animals/The evaporation of water cools organisms and allows them to control their body temperature. [2]

(b) A high specific heat capacity

- Large increase in heat energy results in relatively small rise in temperature of water/ water needs a great amount of energy to increase its temperature and loses a great amount of energy when its temperature decreases;
- Temperature changes within water minimized/ helps living organisms have constant temperature/Cells contain lots of water to keep their temperature constant;
- which is essential for the biochemical processes occurring within their bodies/
- Biochemical processes operate within limits. [2]

.

(c) The cohesive attraction of water molecules to each other.

Individual molecules stick together;

Surface of liquid occupy least possible space;

Allow small organisms to skate on water;

slows down water loss in plants leaves through pores/stomata.
Translocation. [2]
Ref. #1. 6030/2 J2019 NC

3. **(a) Fig 1.1** shows a general water molecule.

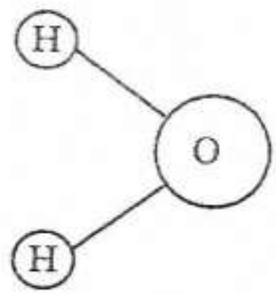


Fig. 1.1

Draw a second water molecule on **Fig 1.1** and show how the two water molecules are bonded together. [3]

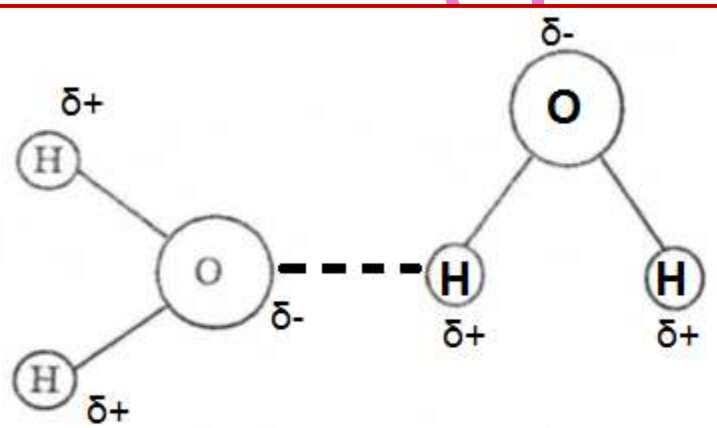


Fig. 1.1

(b) Complete **Table 1.1** by writing the appropriate term for each description.

Table 1.1

Description of reaction	**Term**
Reaction that results in the elimination of water molecules during formation of a bond.	Condensation
The phosphate group of a phospholipid readily attracts water molecules.	Hydrophilic
Splitting of water by light.	Photolysis

[3]

(c) Explain how the following properties of water help organisms to survive in an aquatic environment:

(i) Water acts as a solvent for ions.

- Ions are polar/charged;
- Ions are attracted to/bind to/interact with water;
- Named organisms take up dissolved mineral ions e.g. plants take up nitrates and use them to make proteins. [2]

(ii) A large quantity of energy is required to raise the temperature of water by 1 ^{O}C.

- Many/stable hydrogen bonds between water molecules;
- A lot of energy needed to break the hydrogen bonds;
- Resulting in a high specific heat capacity;
- This means extensive heat absorption with only a small change in temperature [2]

Ref. #1ScA. 9190/2 J2017

4. (a) Fig 2.1 shows the energy changes that occur during an enzyme-catalysed reaction.

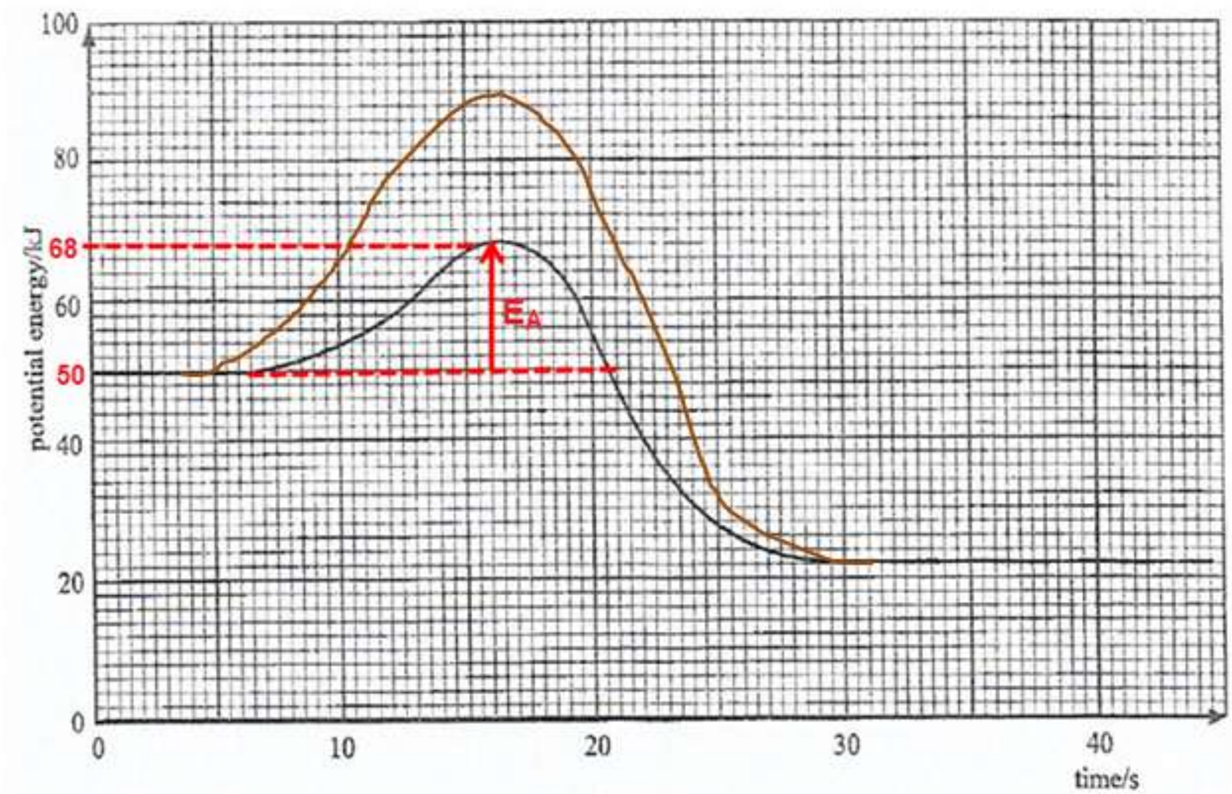


Fig. 2.1

(i) Calculate the activation energy from this reaction.

Maximum PE = 68 kJ

E_A = 68 kJ – 50 kJ

= 18 kJ

Activation energy = **18 kJ** [2]

(ii) Indicate on the curve, the energy changes that would occur in the absence of the enzyme.

Same starting point at 50 kJ with higher curve;

Same ending point at 22 kJ. (see graph drawn above) [2]

(b) Describe any one way in which an enzyme interacts with its substrate.

- Lock and key: Only substrate of complementary shape fits into active site of a particular enzyme
- **OR** Induced fit: Substrate disturbs shape of enzyme and cause it to assume a new configuration which is catalytically active.
- Bonds formed between substrate and active site;
- Formation of ES (enzyme-substrate) complex 4 max [3]

Ref. #2ScA. 9190/2 J2017

5. (a) List any **three** properties of fibrous proteins.

1. Insoluble in water
2. Maintained by hydrogen bonds
3. Long parallel peptide chains
4. Crosslinked at intervals forming long fibres or sheets
5. Chemically stable/unaffected by temperature, pH
6. High tensile strength

ANY THREE [3]

(b) Fig 2.1 shows the structure of a haemoglobin molecule.

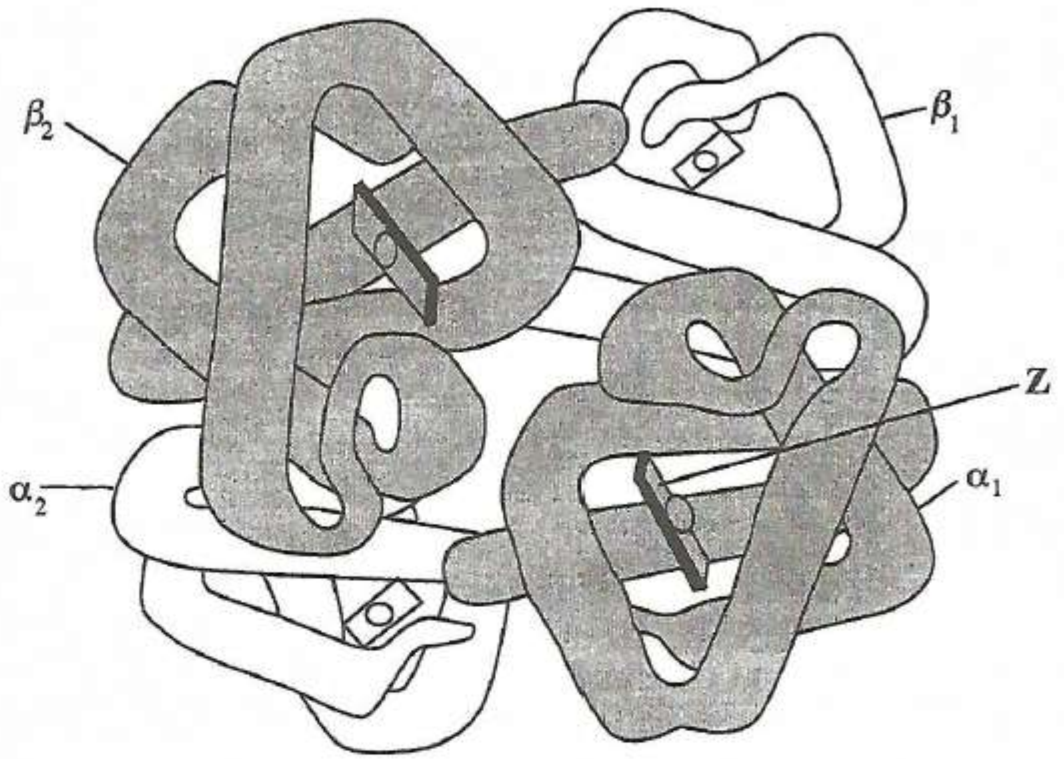


Figure 2.1

(i) Name **Z**.

Haem group [1]

(ii) State the level of organisation for haemoglobin shown in **Fig 2.1**.

Quaternary [1]

(iii) The β chain of haemoglobin has a tertiary structure that consists of eight helices arranged to give a precise 3-dimensional shape.
Describe how the precise 3-dimensional shape of the β polypeptide chain is maintained.

- Held by hydrogen, ionic, disulphide and hydrophobic interactions
- Hydrogen bonds between polar groups
- Ionic bonds between amine and carboxylic acid groups
- Disulphide bonds between cysteines
- Hydrophobic interactions between non-polar/r groups

- Folding occurs in the (primary) structure. 6 max [3]

Ref. #2 ScA 9190/2 N2016

6. **Fig 1.1** shows the structure of haemoglobin molecule, a protein.

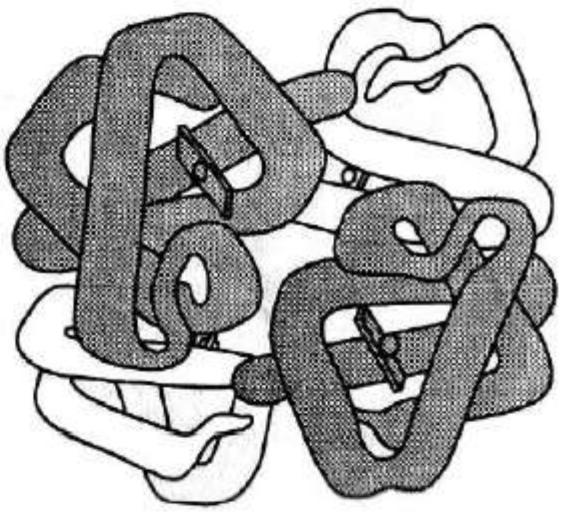


Fig. 1.1

(a) What is meant by the primary structure of a protein?

____

____ [2]

(b) (i) What level of organisation is shown by haemoglobin?

____ [1]

(ii) Describe the globular structure of haemoglobin.

____

____

____

____

____ [4]

(c) Outline briefly how oxygen is carried by haemoglobin.

____

____ [1]

Ref. #1ScA 9190/2 N2004

7. Under conditions of optimum temperature, the enzyme pepsin functions most efficiently over a narrow pH range as shown in **Fig 2.1**.

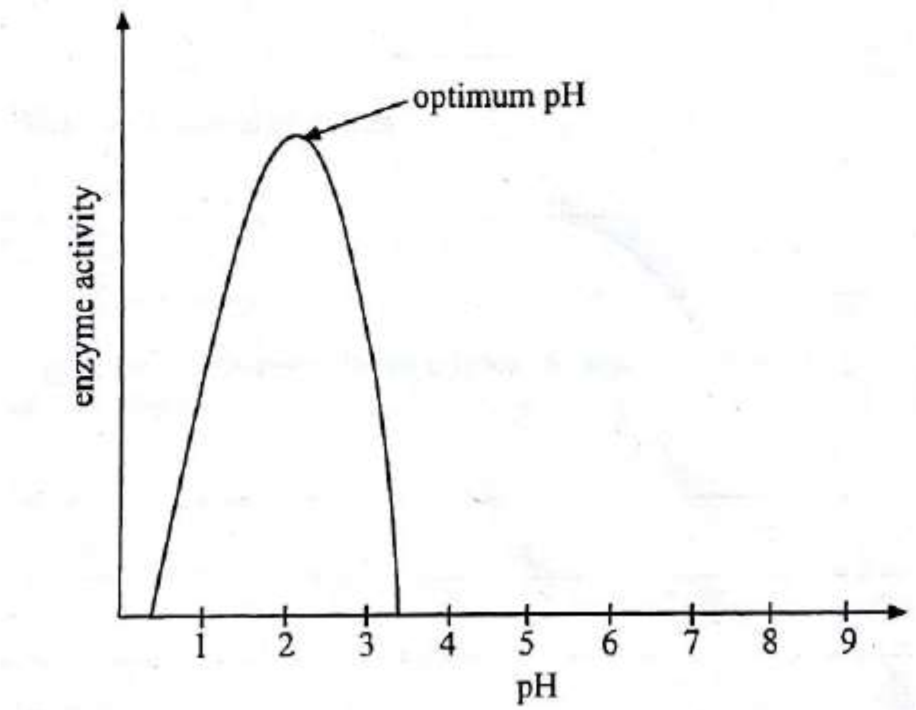


Fig. 2.1

(a) Explain how extreme pH values affect the structure of pepsin.

____

____

____ [2]

(b) Suggest the importance of having different optimum pH values for every biological enzyme.

____

____ [1]

(c) Sketch a graph to show the effect of varying substrate concentration on an enzyme catalysed reaction. [3]

Ref. #2ScA 9190/2 N2004

8. Catalase is an enzyme that catalyses decomposition of the toxic hydrogen peroxide into water and oxygen.
The graph in **Fig 3.1** shows the effect of pH on catalase activity.

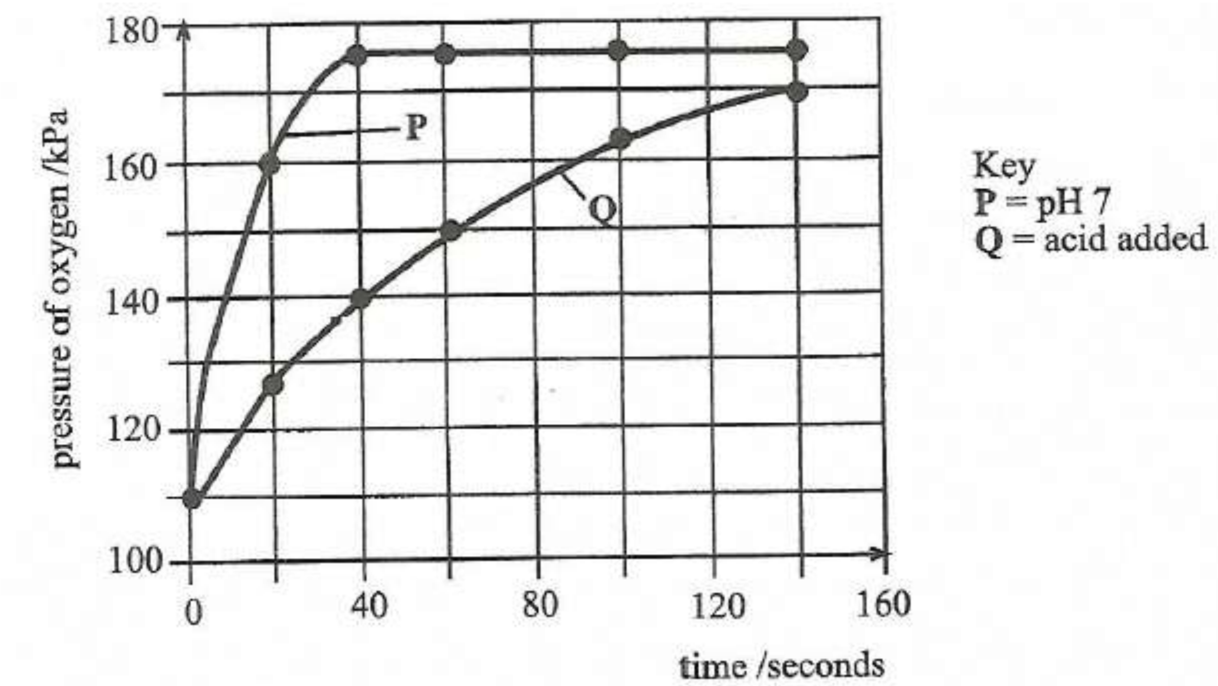


Figure 3.1

(a) **(i)** Describe the shape of graph **P** over time.

Rate was rapid at first; reaching a peak of 175kPa; and then remained constant. [2]

(ii) Explain the shape of graph **P** after 40 seconds.

Hydrogen peroxide was used up/the reaction at maximum rate, Vmax, equilibrium, all active sites occupied. [1]

(iii) Describe the effect of the acid on the activity of catalase enzyme.

- Activity of catalase lowered; (ref to curve Q figures);
- Acid deactivates the enzyme so that the reaction can take place slowly;
- Concentration of H+ ions increases/increase in number of positive charge in medium;
- disrupting ionic bonding that help maintain the specific shape of the enzyme; i.e. the enzyme becomes denatured. [2]

(b) Suggest the mechanism by which hydrogen peroxide may be held in the active site of catalase while the reaction occurs.

- Catalase/enzyme is flexible and moulds/Lock and key –fits precisely.
- Remodels to fit the substrate molecule
- Enzyme substrate complex is formed
- Ref to bonds, ionic, Hydrogen bonds, hydrophobic interactions, van der Waals forces. [3]

Ref. #3 ScA 9190/2 N2016

9. **(a) Fig 3.1** shows results obtained in an investigation on the effect of temperature on the enzyme controlled reaction.

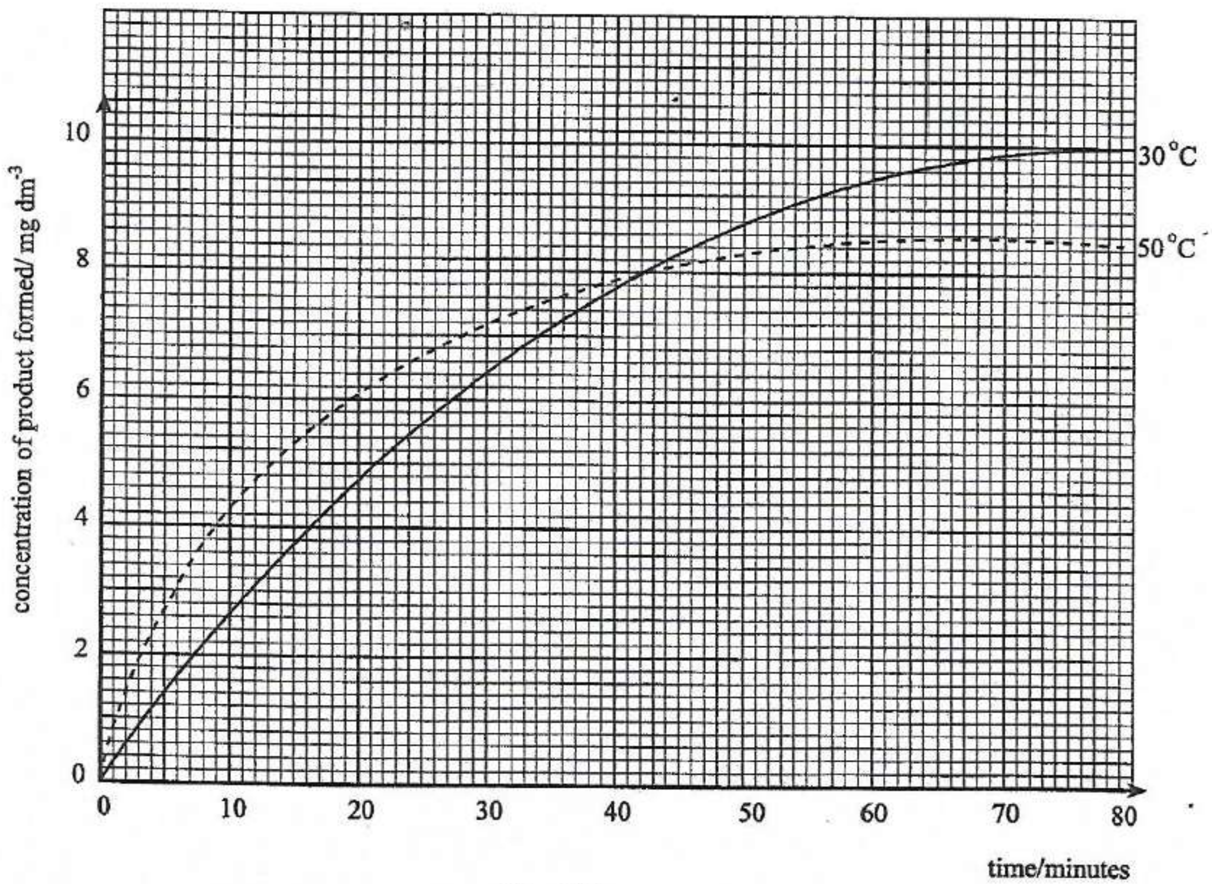


Fig. 3.1

(i) Calculate the rate of reaction in the first 10 minutes at 30°C. Show your working.

$\frac{2.6 \text{ mg dm}^{-3}}{10 \text{ min}} = 0.26 \text{ mg dm}^{-3} \text{ min}^{-1}$

Rate = 0.26 mg dm^{-3} min^{-1} [2]

(ii) Explain why the rate of reaction was faster at 50°C than at 30°C during the first 20 minutes.

Higher kinetic energy; more collisions; leading to faster and more formation of E-S complexes [2]

(b) Explain the shape of the curve from 20 to 60 minutes at 50°C.

Breaking of hydrogen bonds ; leading to change in structure of active site; Denaturation of enzyme. substrate no longer binds/fits in active site. [3]

Ref. #3ScA. 9190/2 N2015

10. (a) Distinguish between the structure and function of a named globular protein and a named fibrous protein.

1. Haemoglobin (Hb)/enzyme/Myoglobin (Mb) globular;
2. Collagen/keratin fibrous;
3. Polypeptide chains with irregular sequence of amino acids;
   Collagen regular repetitive sequence of amino acids/four polypeptides for Hb and three for collagen;
4. Hb shape is a compact globule of polypeptides collagen has long chains running parallel;
5. Hb chemically less stable whilst collagen chemically stable;
6. Hb has haem prosthetic group whilst collagen does not have;
7. Globular protein is water soluble whilst fibrous is insoluble in water;
8. Globular involved in various body systems such as digestive/transport/endocrine/immune system; fibrous helps to maintain structure and provide support;
9. Globular-tertiary structure/quaternary most important whilst in fibrous secondary is more important/bonds involved [8]

(b) Describe the types of bonds which help to hold protein molecules in shape.

Ionic bond

- Formed when amino acids carrying opposing electrical charges are next to the hydrophilic area of proteins;
- Occur on exterior of protein;
- Provides electrostatic attractions;
- Form when H+ ions transfer from the carboxyl group of one amino acid to the amino group of another.

Hydrogen bonds

- Formed when two electronegative atoms interact with the same hydrogen;
- Contribute to stability;
- Forms secondary structure, such as alpha helix/beta sheet;

Hydrophobic bonds

- Cause folding of globular proteins;
- Non-polar groups (that repel water);

Disulphide bonds

- Formed between oxidation of sulf hydryl groups (on cysteine) between two sulphur atoms;
- Strong covalent bonds that hold protein chains or loops;
- Hold catalytic site of enzyme; [8]

Ref. #10ScB. 9190/2 N2015

11. Fig 2.1 shows some types of bonds which stabilize protein structure.

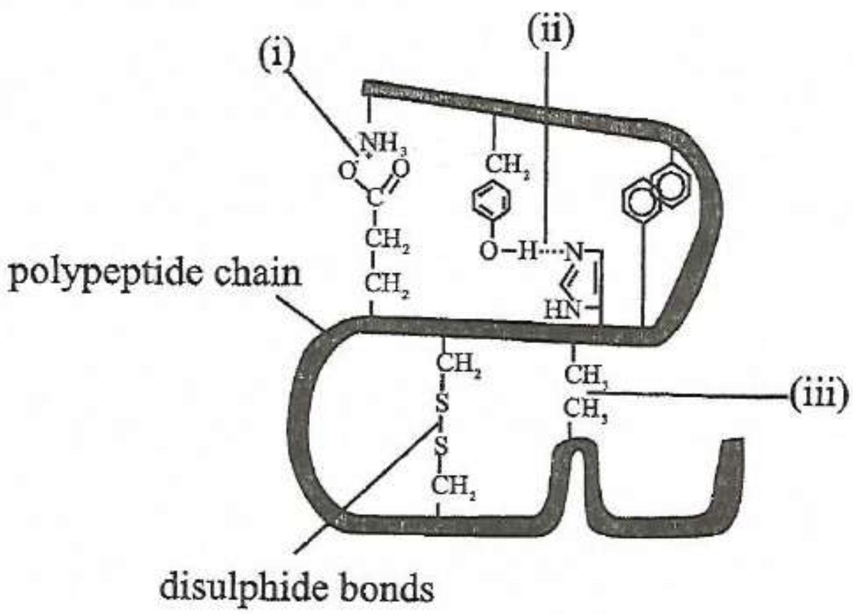


Fig. 2.1

(a) Identify the bonds marked **(i), (ii)** and **(iii).**

i. Ionic bond
ii. Hydrogen bond
iii. Hydrophobic interaction [3]

(b) State the level of protein structure maintained by the bonds in **Fig 2.1**.

Tertiary structure [1]

(c) Explain what determines the precise position of amino acids in the structure of a polypeptide chain.

Base sequence of DNA;

Precise sequence of nucleotides on mRNA [2]

Ref. #2ScA. 9190/2 N2014

12. The catalytic cycle of the hydrolysis of sucrose involves the binding of the sucrose to the active site of invertase to form a complex structure.

(a) Name the **two** monosaccharides produced from the hydrolysis of sucrose.

1 Glucose
2 fructose [2]

(b) Describe how the sucrose may be attached to the active site of the enzyme invertase.

Hydrogen bonds (between the sucrose and amino acids located on the active site)/ionic interactions/hydrophilic interactions;
Lock and key/induced fit. [2]

(c) Explain how the induced fit hypothesis can be used to explain how the sucrose is hydrolysed by invertase.

- Active site changes shape when sucrose binds;
- Enzyme substrate complex is formed;
- Bonds are strained between substrate and active site; thereby lowering the activation energy (E_A)
- Enzyme-substrate (E-S) complex breaks to give enzyme and products. [4]

Ref. #3ScA. 9190/2 N2014

13. The enzyme urease catalyses the following reaction,

Urea + Water $\longrightarrow$ **Ammonium carbonate**.

Ammonium carbonate readily gives off ammonia.

The indicator bromothymol blue is yellow in neutral solution and blue in alkaline solution. In an investigation urease was mixed with bromothymol blue. The mixture was divided equally among five test tubes labelled **A** to **E**.

One test tube was placed in each of the five water baths at the temperatures shown in **Table 2**. The same mass of urea was added to each test tube. The time taken for the contents of each test tube to turn blue was recorded. The results of the investigation are shown in **Table 2.**

Table 2

Test tube	Temperature/°C	Time taken for blue colour to appear /s
A	0	89
B	15	21
C	35	5
D	45	17
E	55	33

(a) Suggest why the indicator changed colour after urea was added to the enzyme solution.
Production of ammonium carbonate/ammonia [1]

(b) With reference to **Table 2**,
(i) state what can be concluded about the optimum temperature of the enzyme.
Urease optimum temperature is $35^{O}C$ [1]

(ii) Explain why the time taken for the blue colour to appear is greater

1. at $15^{o}C$ than at $35\ ^{o}C$,
   at $15^{O}C$, time taken is greater because the enzyme is still less active/less kinetic energy/less collisions;
   hence less enzyme-substrate complexes are formed/ (due to less effective collisions) [2]

2. at $55^{o}C$ than at $35\ ^{o}C$.
   at $55\ ^{O}C$, time taken is greater because enzyme urease is being denatured by excess heat;
   hence less active sites /enzymes are available for reaction. [2]

(c) The indicator was added to a separate sample of urea and to a separate sample of urease. In neither test tube was a blue colour produced.
Explain why the indicator was added to separate samples of both urea and urease.
To show that the colour was produced as a result of hydrolysis of urea by urease;
Urea or urease alone would not produce an alkaline environment in the tube. [2]
Ref. #2ScA. 9190/2 N2013

14. (a) Describe the biological significance of the following properties of water:
(i) High heat capacity
Amount of heat required to raise temperature of 1kg of water by $1^{O}C$;
Large increase in heat energy results in relatively small rise in temperature of water;
Temperature changes within water minimized;
Biochemical processes operate within limits. 4 max [2]

(ii) High surface tension and cohesion
Individual molecules stick together;
Surface of liquid occupy least possible space;
Allow small organisms to skate on water;

Translocation. 4 max [2]

(iii) Universal solvent.

Water is polar molecule;

Ionic substances dissolve;

Ions/substances move freely;

Acts as a transport medium [2]

(b) Explain the functions of water to living organisms.

- Reactant for photosynthesis/Hydrolysis of substances;
- Solvent;
- Support in aquatic organism/Turgidity in plant cells/penis erection/hydrostatic skeleton;
- Medium for diffusion;
- Fertilisation in swimming gametes;
- Translocation/transpiration;
- Lubrication;
- Germination;
- Protection-mucus/tears max [6]

Ref. #9ScB. 9190/2 N2013

15. Fig. 1 shows the structure of glycerol.

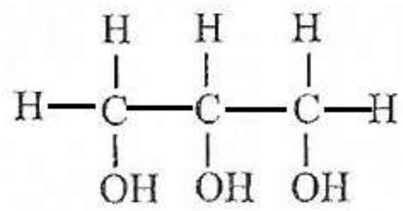


Fig. 1

Some molecules are produced by condensation reactions between glycerol and other compounds. The products of these reactions may have structural or storage roles in cells.

(a) Name a molecule, which is produced in this way, which has:

(i) a storage role, triglyceride/Fat or oil

(ii) a structural role, phospholipid; glycolipid [2]

(iii) Glycerol molecules diffuses into cells more rapidly than glucose molecules. Suggest a reason for this.

Glycerol molecules are smaller/half the size of glucose molecules;

Glycerol =3C whilst glucose = 6C;

Transport proteins maybe more plentiful for glycerol. [1]

(b) In cells, more than one pathway may lead to the synthesis of glycerol.
Describe, using an example, what is meant by a *metabolic pathway*.

A series of chemical reactions in which the product of one reaction becomes the substrate/starting point for the next reaction. Is usually controlled/catalysed by a specific enzyme.

ANY NAMED EXAMPLE, e.g. glycolysis/photosynthesis [2]

Ref. #1ScA. 9190/2 N2011

16. Describe the structure of a named globular protein.

ANY CORRECT NAMED EXAMPLE e.g. enzyme, haemoglobin
Polypeptide chains have tertiary/quaternary structure
Fold into compact 3D shape
Each protein has its own specific shape and length of chains; ACCEPT specific description;
Soluble in water, making colloidal solutions;
Changed chemically and so not chemically stable;
Irregular folding formed by bonds between R-groups/bonds to R groups. [6]

Ref. #11(a)ScB. 9190/2 N2011

17. (a) Fig 2.1 shows a three dimensional representation of the structure of lysozyme, an enzyme found in saliva and tears, as well as other body fluids.

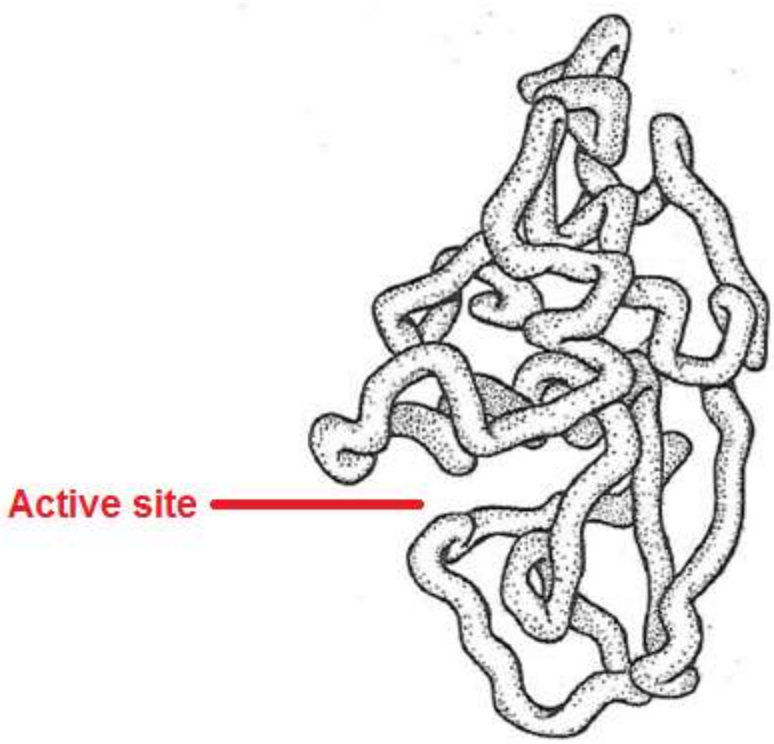


Fig.2.1

(i) On **Fig 2.1** label the *active site*. [1] **ACCEPT** any labelled groove

(ii) Describe the role of the *active site*.

Site where enzyme binds; to substrate/cell wall; by lock and key mechanism/induced fit theory. Substrate/ bacterial cell wall broken down (into sub units/products/smaller pices). 4 max [2]

(iii) State the group of biological molecules to which lysozyme belongs.

Protein [1]

(iv) Describe the roles of lysozyme in tears and saliva.

When pathogens/bacteria enter the eye/mouth; lysosomes break the bacterial walls; killing the bacteria. [3]

(b) Explain why enzymes are effective in very small quantities.

Not used up in reaction/can be used again and again; high turn over/often work fast [2]

Ref. #2ScA. 9190/2 J2015

18. **(a)** **Table 3.1** summarises differences between starch, glycogen and cellulose. Complete **Table 3.1**

	Starch	Glycogen	Cellulose
Name of monomer	Glucose	Glucose	β-glucose
Bonds between monomers	1,4 glycosidic (amylose). 1,4 and 1.6 glycosidic (amylopectin).	1,4 and 1,6 glycosidic	1,4 glycosidic
Characteristics of chains	Coiled unbranched and long branched	Short branched chains	Straight long unbranched chains

[3]

(b) (i) Some small insects are able to stand on the surface of water.
State the property of water that enables insects to stand on it.
High surface tension [1]

(ii) Outline the importance of high heat of vaporization of water in living organisms.
High amounts of heat energy are needed to overcome hydrogen bonds to vaporize a small quantity of water.
Makes water very efficient in cooling process, e.g. transpiration in plants and sweating in animals/The evaporation of water cools organisms and allows them to control their body temperature. [2]

(c) Explain why cell contents and their environment are less likely to freeze under extremely cold temperatures.

Due to high heat capacity; high amount of heat is needed to break hydrogen bonds in water; which restrict movement of molecules/boiling; Water must loose large amounts of heat energy to freeze 4 max [3]

Ref. #3ScA. 9190/2 J2015

19. Fig 2.1 shows the structure of two biological molecules **A** and **B**.

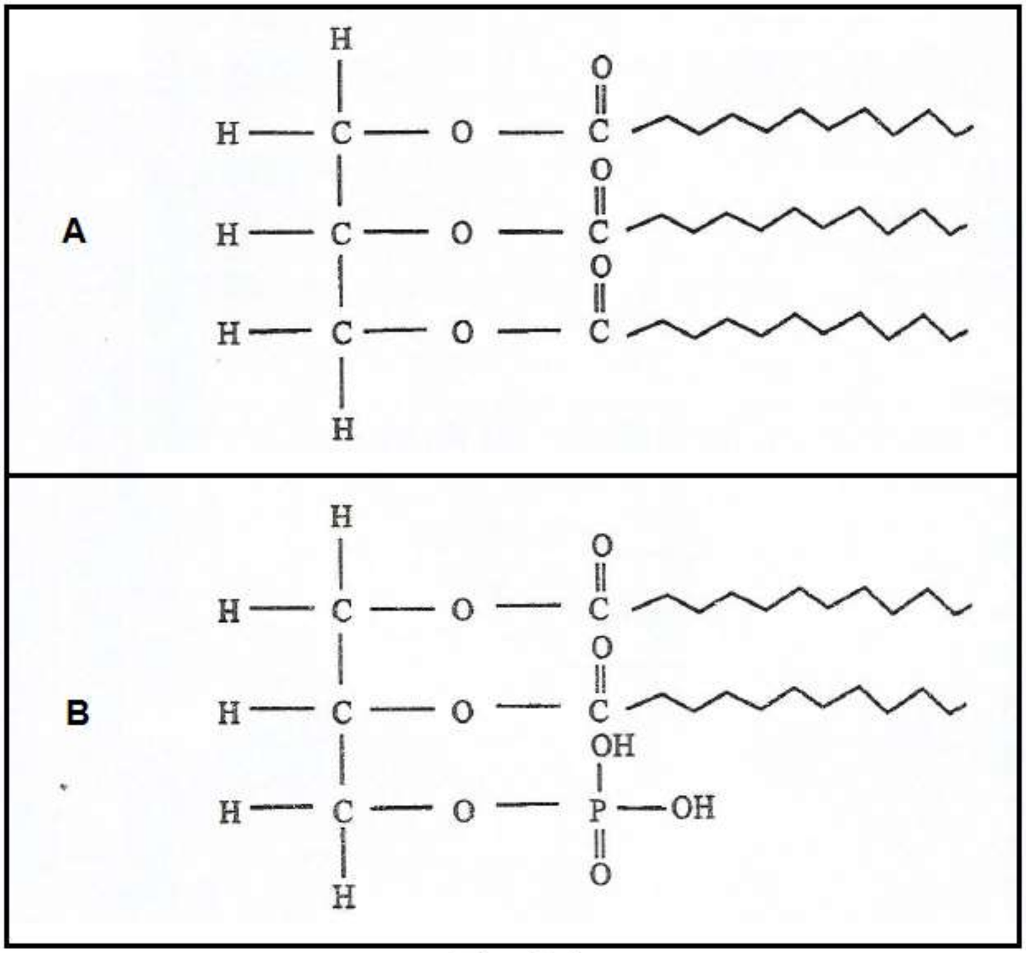


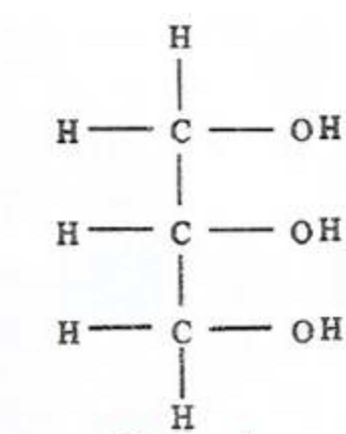
Fig. 2.1

(a) (i) Name the molecules **A** and **B**.

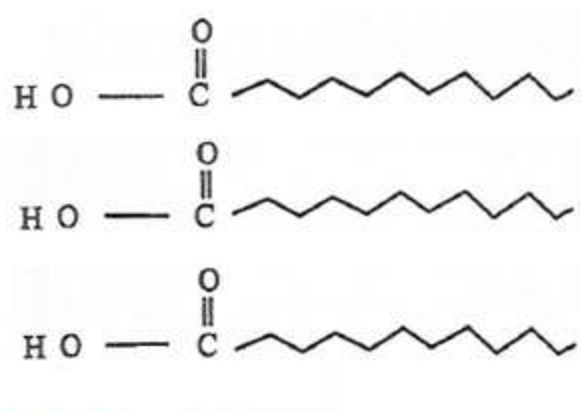
A. Triglyceride

B. Phospholipid [2]

(ii) Draw the structural formulae of the products of the hydrolysis of molecule **A.**



Glycerol



3 Fatty acids

[2]

(b) Explain the significance of double bonds in the hydrocarbon tails of some fatty acids.

Cause kinks in tail; thereby preventing close packing.

Lipids melt easily/have low melting point;

Reducing accumulation of fats in blood vessels. [2]

(c) Explain the importance of poikilothermic (cold blooded) animal cells having a higher proportion on unsaturated fatty acids than homeothermic (warm blooded) animal cells.

Body temperature of cold blooded animals becomes lower in cold environments;

Unsaturated fatty acids have low melting points;

So remain liquid/fluid at temperatures lower than 5°C;

So that cell membrane continues to function normally (under low temperatures).

4 max [3]

Ref. #2ScA. 9190/2 J2016

20. The graph in **Fig 3.1** shows the energy changes of a chemical reaction without enzyme.

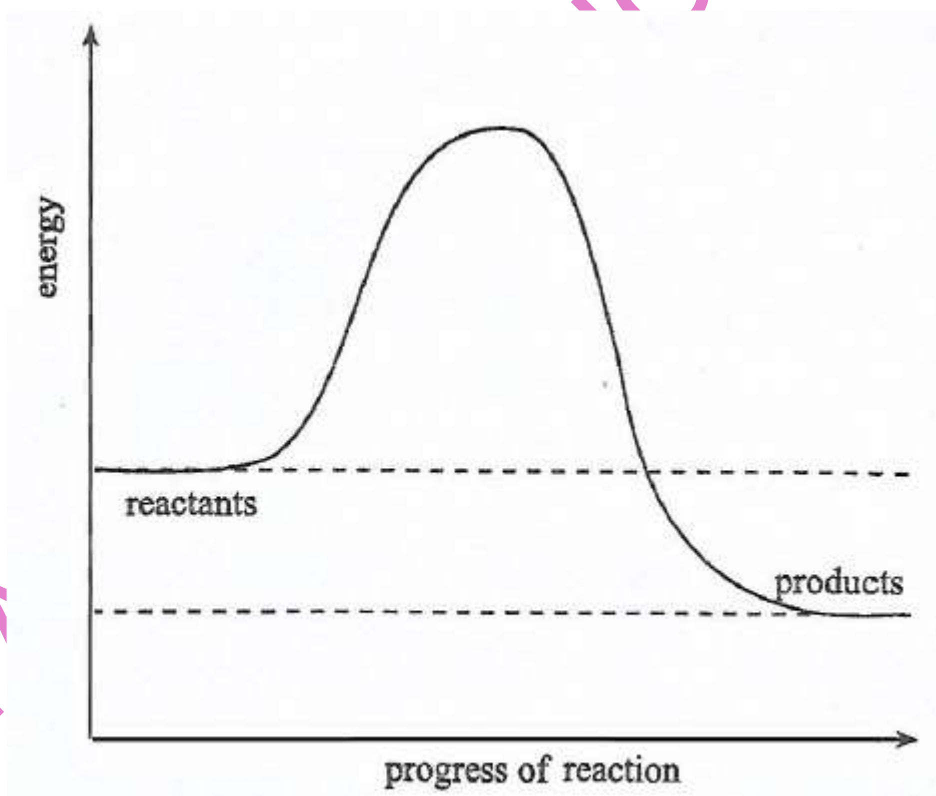
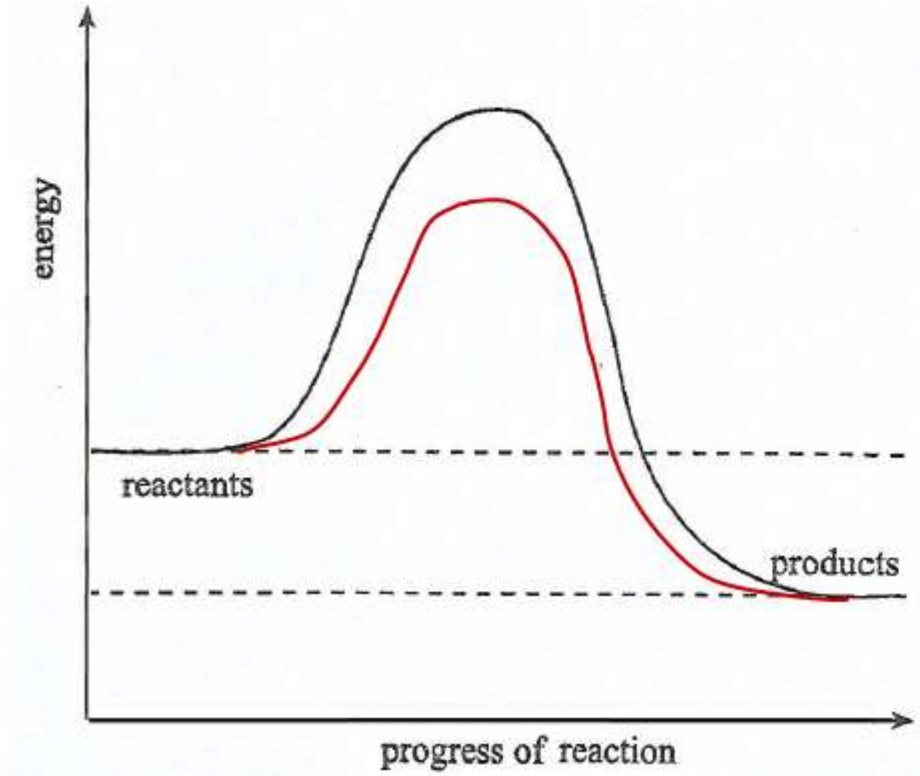


Fig. 3.1

(a) (i) Define the term *activation energy*.

Minimum quantity of energy that must be provided; to start a reaction [2]

(ii) Draw, on **Fig 3.1**, a reaction profile for the enzyme catalysed reaction.



Graph drawn above substrate/reactants line but below graph for activation without enzyme [1]

(b) Describe how an enzyme catalyses a reaction.

Substrate binds to active site;

Its shape slightly changed;

Substrate easily changed into products /makes it easier for bonds to be broken.

[3]

(c) Heavy metal ions such as mercury, silver and arsenic inhibit enzymes.

(i) Name the type of inhibition caused by the heavy metals on enzymes.

Non-competitive [1]

(ii) Describe how heavy metals cause the inhibition in **c(i)**.

Bind permanent to sulphydryl/disulphide bonds;

Inactive site/elsewhere;

Causing change in enzyme structure;

Enzyme protein precipitate 4 Max [3]

Ref. #3ScA. 9190/2 J2016

21. Fig 2.1 shows the general formula of an amino acid.

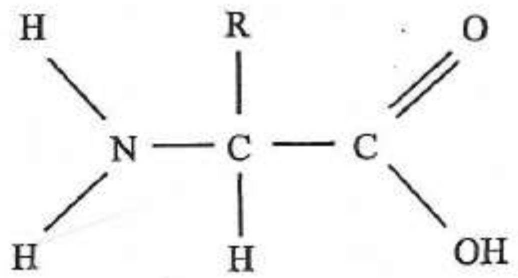


Fig. 2.1

(a) Amino acids form bonds when making up a polypeptide. Identify the following bonds.

(i) $\vdash CO—HN \dashv$ = ______________________________

(ii) $\vdash COO^{-}\quad H_3\overset{+}{N}— \dashv$ = ______________________________

(iii) $\vdash OH \cdots\cdots OC — \dashv$ = ______________________________

[3]

(b) **(i)** Using **Fig 2.1** identify the part which is involved in hydrophobic interactions.
____ [1]

(ii) Which bond between amino acids is a result of

1. condensation reaction,
____ [1]

2. pH interaction?
____ [1]

(c) Describe how you would recognise the presence of the bond in (a)(i) in a substance.
____
____ [2]

Ref. #2ScA. 9190/2 N2005 Silver jubilee

22. **(a)** Using catalase as an example, describe an experiment to study the effect of different temperatures on the course of an enzyme catalysed reaction. [8]

(b) Explain the effect of reversible inhibitors on the rate of an enzyme activity. [8]

Ref. #9ScB. 9190/2 N2005 Silver jubilee

23. **Fig 2.1** shows molecules of enzymes, substrates and inhibitors about to react.

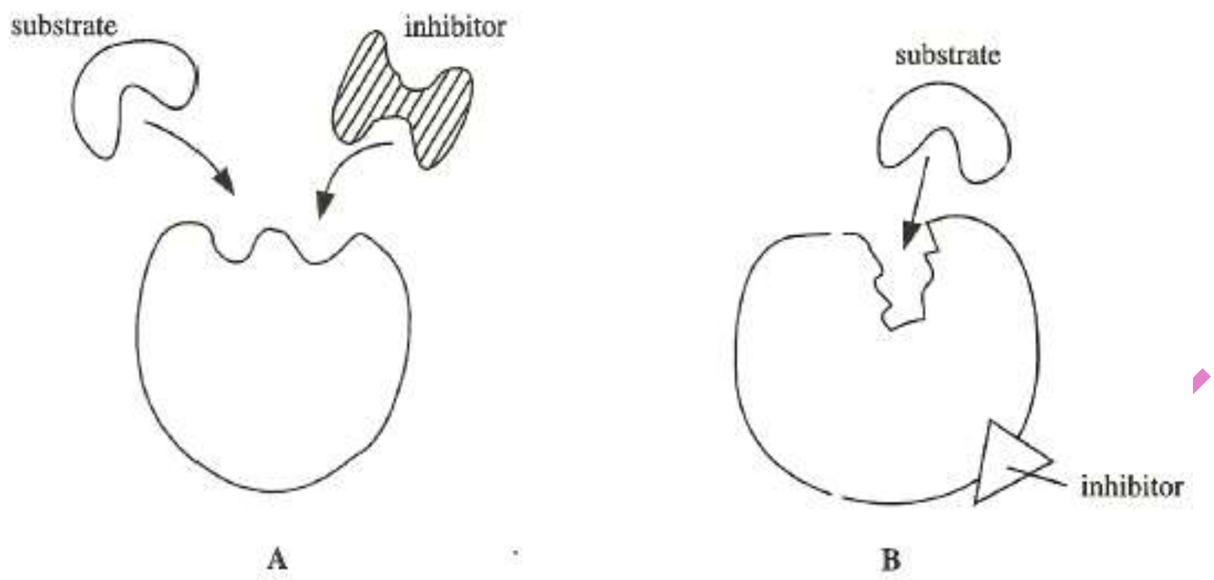


Fig. 2.1

(a) State the type of inhibition shown by **A** and **B**.

A. ____

B. ____ [2]

(b) With reference to the diagrams above describe the differences between the types of inhibition stated in **(a).**

____

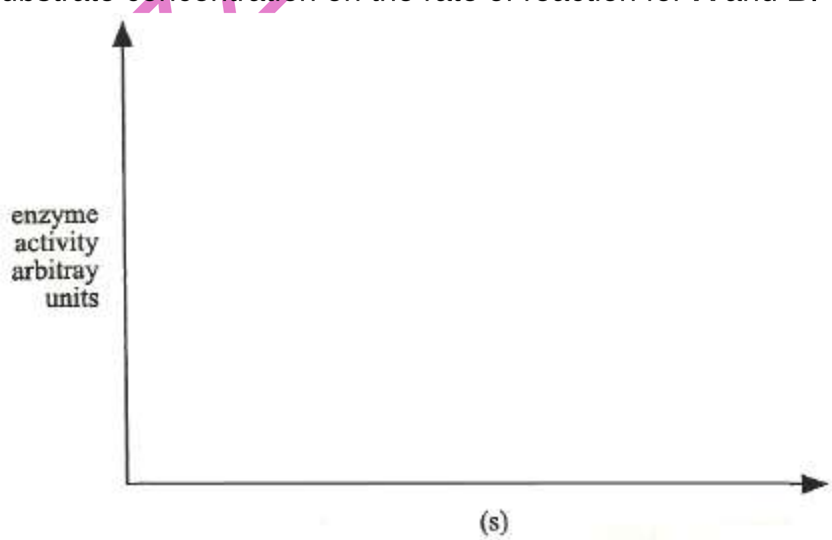
____

____

____

____ [3]

(c) On the axes below sketch a graph to show the effects of an increase in substrate concentration on the rate of reaction for **A** and **B**.



[2]

Ref. #2ScA. 9190/2 J2006

24. (a) Describe how the properties of triglycerides are related to their functions in living organisms. [5]

(b) Explain the roles of water in living organisms. [7]

Ref. #10ScB. 9190/2 J2006

25. (a) (i) Triglycerides are biological molecules of great importance to living organisms. In the space below, draw a labelled diagram to show the general structure of a triglyceride. [2]

(b) Explain why triglycerides are insoluble in water.

____

____

____

____ [3]

(c) Suggest why triglycerides release more energy on oxidation compared to an equivalent mass of carbohydrates.

____

____

____ [2]

Ref. #1ScA. 9190/2 N2008

26. Cylinders of a potato were made from a tuber with a cork borer. The cylinders were then sliced into discs of the same thickness and put into a small beaker containing 50 cm3 of 2 Molar hydrogen peroxide. A reaction took place which produced water and oxygen. The mass of the beaker and its contents were recorded over a period of 15 minutes. The results are shown in **Fig 2.1**.

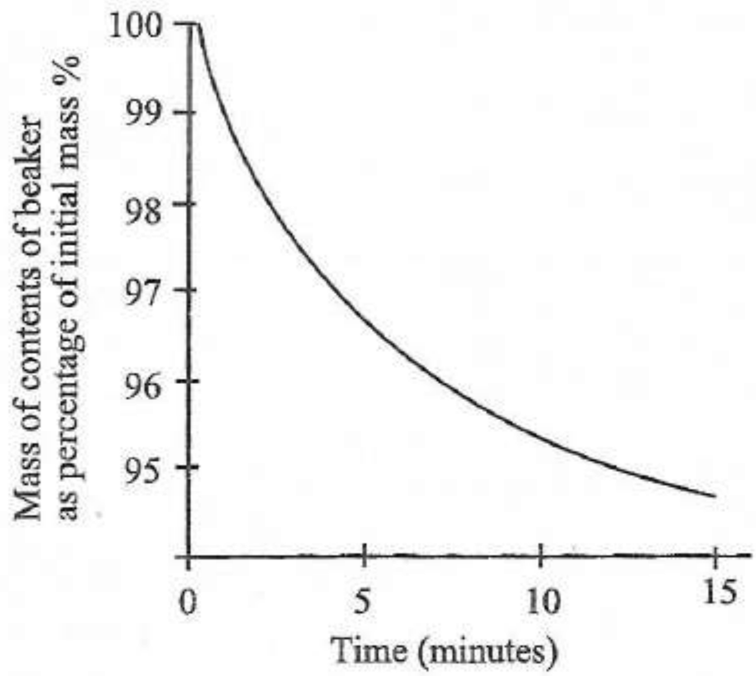


Fig. 2.1

(a) What caused the reaction to take place?

____

____ [1]

(b) State two other factors which should be maintained at a constant level when conducting the experiment.

____

____ [2]

(c) Explain why the mass of the contents of the beaker fell as the reaction progressed.

____

____

____

____[2]

(d) Account for the change in the rate of reaction over the period of time shown on the graph.

____

____

____

____[2]

Ref. #2ScA 9190/2 N2008

27. Table 1.1 shows the number of carbon atoms contained in some substances.

Table 1.1

Substance	Number of carbon atoms
Amino acid	2 – 11
Glycerol	____
Glucose	____
Starch	Large and variable number

(a) Complete the table to show the number of carbon atoms in glycerol and glucose. [2]

(b) Explain why the number of carbon atoms

(i) may vary from 2 to 11 in an amino acid molecule;

this depends on the nature of the fourth group attached to the 2-carbon atom/the composition of the R-group. [1]

(ii) may be described as variable in a starch molecule.

Because the proportions of amylose and amylopectin vary. [1]

(c) Explain why the type of chemical reaction in which glucose residues are joined together to from starch is referred to as condensation.

Because a water molecule is released; each time a glycosidic bond is formed. [2]

Ref. #2ScA 9190/2 N2009

28. **(a)** Describe the molecular structure of an enzyme.

Complex 3-dimensional globular protein;

Have specific sequences of amino acids;

A few catalytic amino acids making up a region called the active site;

Which reacts with substrate;

Soluble/relatively stable;

Tertiary structure maintained by disulphide, ionic and hydrogen bonds. [5]

(b) Explain the effects of temperature on enzyme activity.

- As temperature increases reaction rate increases;
- Due to the increased kinetic energy of the substance and enzyme molecules;
- This is until an optimum temperature is reached;
- Enzymes are denatured beyond this point;
- by breaking hydrogen bonds and other forces holding the enzyme molecule in their precise 3-dimensional shape.
- Active site no longer fit substrate;

- *Rate of activity decreases.* [7]

Ref. #10ScA. 9190/2 N2009

TOPIC 3 CELL AND NUCLEAR DIVISION

1. In an investigation a student observed cells in a stained section of Meristematic tissue.

(a) (i) Name any **two** parts of a plant where Meristematic tissue are found.

1 ____

2 ____ [2]

(iv) Explain the purpose of a stain in the investigation.

____

____ [1]

(b) The student counted the cells seen in each stage of the cell cycle.

Table 3.1 shows the results.

Table 3.1

Stage of cycle	Percentage cells in stage (%)
Interphase	82.00
Prophase	4.34
Metaphase	3.23
Anaphase	3.23
Telophase	7.20

Calculate the percentage of the cell cycle taken up by nuclear division. [2]

(c) State any one way in which the products of meiosis are different from the products of mitosis.

____

____ [1]

Ref. #3. 6030/2 J2019 NC

2. Fig 3.1 shows chromosomes at prophase 1 of meiosis.

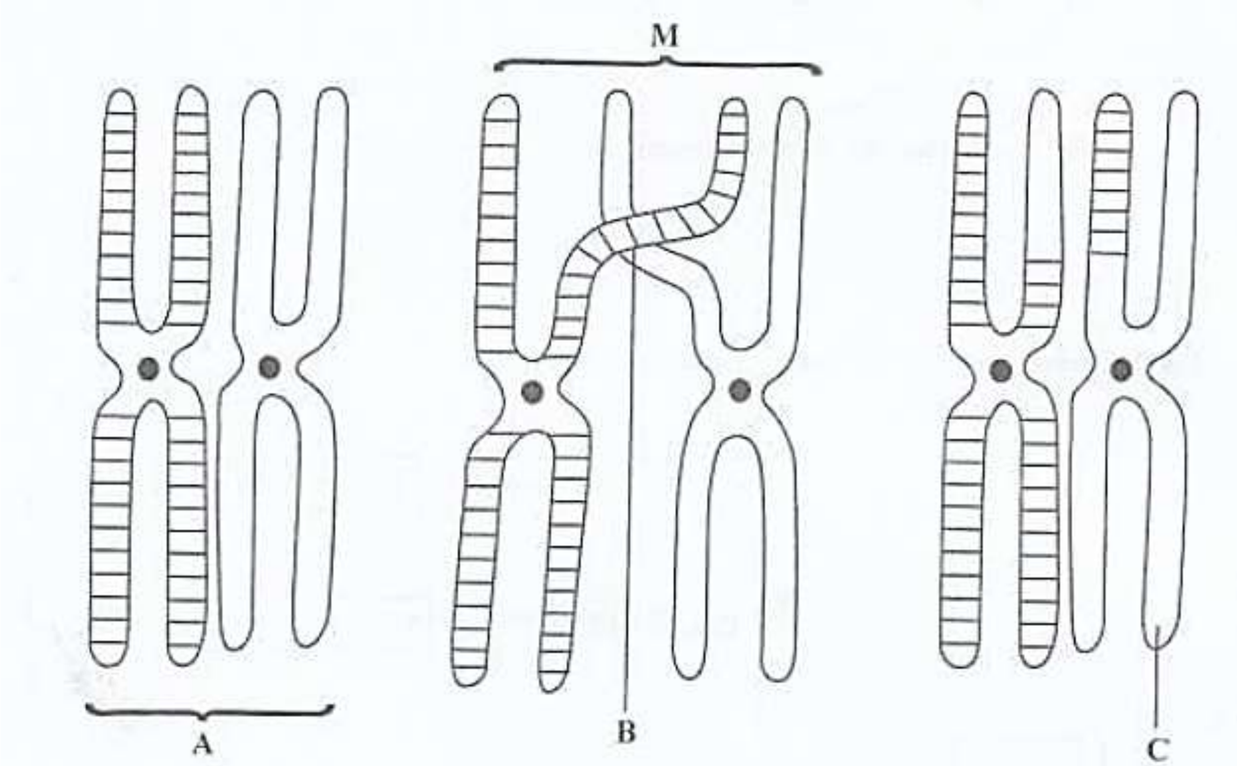


Fig.3.1

(a) Identify the process:

1. which resulted in pairing of the chromosomes

____

2. occuring at chromosome pair **M**

____

(b) Name the structures labelled **A**, **B** and **C**.

A ____

B ____

C ____ [3]

(c) Calculate the possible number of chromosomal pair combinations in a cell in which 2n = 46. [2]

Ref. #3ScA. 9190/2 J2018

3. **(a) Fig 3.1** shows the stages in the development of cancer.

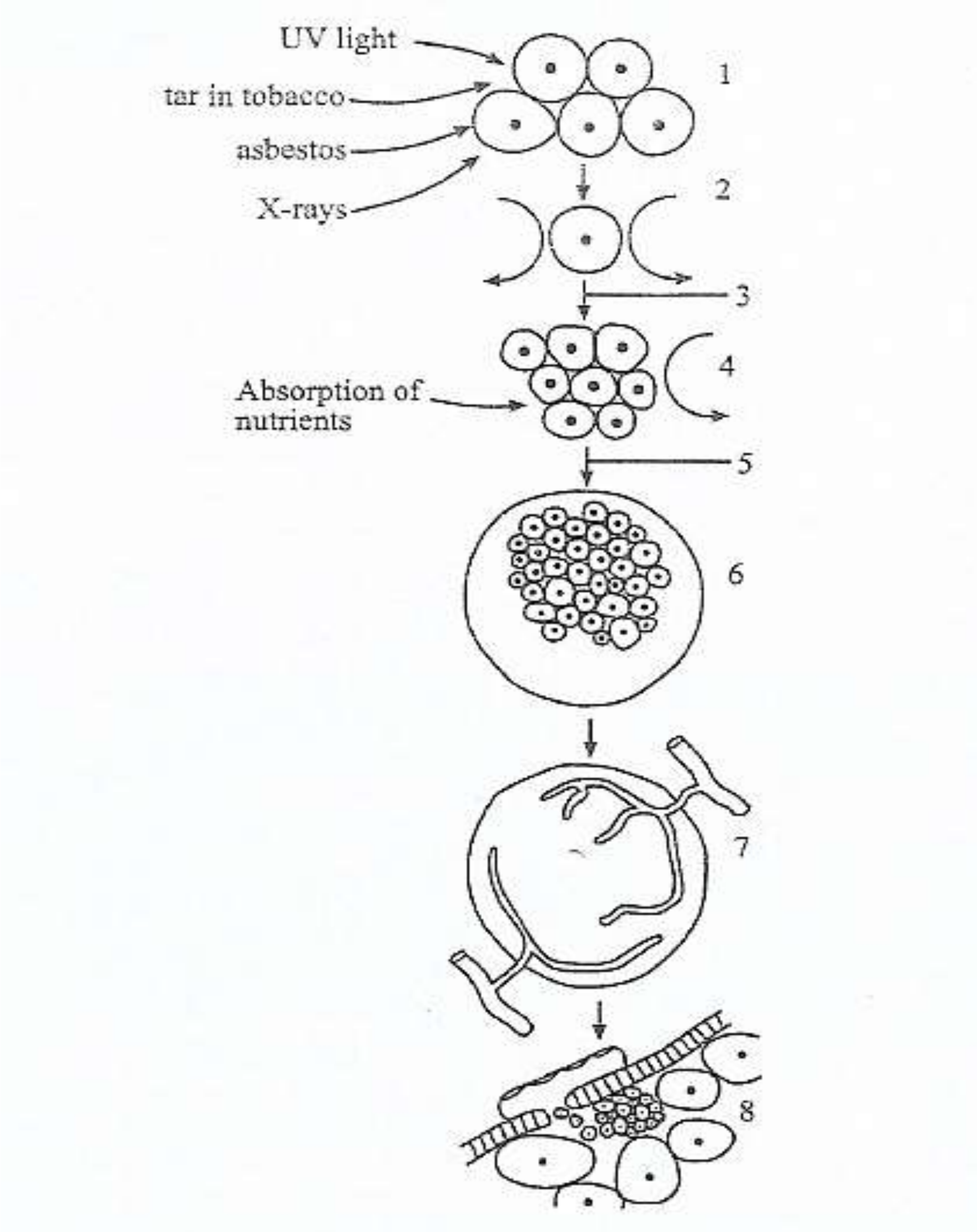


Fig. 3.1

Identify stages 3, 6 and 8 on **Fig 3.1.**

3 : Mitosis

6 : Tumour

8 : Metastasis. [3]

(b) Distinguish between benign and malignant tumors.

Benign tumors are localized whilst malignant tumors exhibit metastasis (spread throughout the body)/Benign tumors replace surrounding tissue whilst malignant tumors invade other tissues. [2]

(c) Explain why cancer is difficult to treat when it has reached the secondary stage.
Because it has spread throughout the body/has metastasized; and has become malignant/has invaded other tissues. [2]
Ref. #3ScA. 9190/2 J2017

4. (a) Describe the behaviour of chromosomes during meiosis which results in genetic variation.

- Homologous chromosomes pair up/synapsis;
- during prophase I;
- to form bivalents;
- crossing over;
- chiasma(ta) occur;
- DNA/alleles exchanged/new gene combinations
- Independent assortment; of homologous chromosomes;
- during metaphase I;
- bivalents line up on equator, independent of each other/randomly;
- independent assortment of chromatids;
- during metaphase II;
- AVP e.g. non disjunction/separation during Anaphase I. 12 max [8]

(b) Explain the significance of mitosis.

- Maintains genetic stability in a species;
- by ensuring same number of chromosomes in all daughter cells;
- daughter cells genetically identical to mother cells;
- basis for growth in multicellular organisms; by increase in cell number;
- Replacement of worn out tissues;
- Regeneration of body parts e.g. arms in starfish/legs in crustaceans.
- Basis for asexual reproduction; e.g. binary fission/budding. [8]

Ref. #9ScB. 9190/2 N2016

5. Outline the differences between mitosis and meiosis.

	Mitosis	Meiosis
DNA replication	1. Occurs during interphase before mitosis begins	Occurs during interphase before meiosis I begins
Number of divisions	2. One, (including prophase, metaphase, anaphase and telophase)	Two, (each including prophase, metaphase, anaphase and telophase)/has two divisions:-1st = reduction to 2 haploid cells. -2nd =reduction to 4 haploid cells.
Prophase	3. homologous chromosomes remain separate	Homologues pair up
	4. No formation of chiasmata	Chiasmata form
Metaphase	5. Pairs of chromatids line up on spindle equator	Pairs of chromosomes line up on spindle equator
Anaphase	6. Centrioles divide.	Centrioles do not divide
	7. Chromatids separate	Chromosomes separate
	8. Chromatids identical	Chromatids may not be identical
Telophase	9. Same number of chromosomes in daughter cells as in parent cells	Half number of chromosomes in daughter cells
Where it occurs	10. Occurs in somatic cells/May occur in haploid, diploid or polyploidy cells	Only occurs in diploid cells/Occurs in reproductive organs – animals (testes/ovaries), -plants (anthers/ovaries)
Cells produced	11. Two diploid daughter cells	Four haploid daughter cells
	12. Genetically identical to parent cells	Genetically different from parent cells
Role in the body	13. Enables multicellular adult to arise from zygote; produces cells for growth, repair and asexual reproduction.	Produces gametes; reduces number of chromosomes by half; introduces genetic variability among the gametes;

13 max [8]

Ref. #12(a)ScB. 9190/2 N2014

6. Some cells divide by mitosis.

(a) (i) Describe the product of mitosis.

- Two daughter cells produced (from one parent cell);
- Have the same number of chromosome as parent cells/genetically identical material;
- Have the same type of cells as parent cells/cells identical to parent cells;
- Cells identical to each other; [2]

(ii) State two processes in which mitosis occurs in humans.

- Growth;
- Repair/replacement of damaged cells or dead cells/regeneration [2]

(b) The diploid number of chromosomes of a gorilla is 48:
State the number of chromosomes which would be found in the following cells of the gorilla,

(i) brain cell: 48

(ii) sperm: 24 [2]

(c) Cancer is caused by uncontrolled mitotic division of cells.
State **three** factors which increase the chances of cancerous growth.

1. Carcinogens/Named carcinogens e.g. ionizing radiation, tar in tobacco smoke, asbestos dust, X-rays etc
2. Age; Viruses/Named example e.g. human papilloma virus (HPV)
3. Genetic factors/hereditary predisposition/genes inherited from parents
4. Mutations of proto-oncogenes /tumour suppressor genes. 4 max [3]

Ref. #3ScA. 9190/2 N2013

7. (a) Explain how DNA replicates semi-conservatively during interphase.

1. Helicase unwinds and separates the double-stranded DNA. To create a replication fork of two antiparallel strands.
2. DNA gyrase relieves strain created by the unwinding of DNA by helicase.
3. SSB proteins (single stranded binding proteins) prevents separated strands from re-annealing/SSB proteins prevent the single stranded DNA from being digested by nucleases.
4. DNA Primase generates a short RNA sequence (called a primer) to initiate DNA synthesis.
5. DNA polymerase III extends new strand in the 5'→3' direction by joining nucleotides together using energy from hydrolysis (breaking down) of dNTP.
6. DNA polymerases also proofread newly made DNA, replacing any incorrect nucleotides. This ensures that each new DNA double helix is exactly like the original.
7. The leading strand is synthesised continuously towards the replication fork.
8. The lagging strand is synthesised away from the replication fork in short pieces called Okazaki fragments.
9. DNA polymerase I remove RNA primers on lagging strand and replace them with DNA nucleotides.
10. DNA ligase joins Okazaki fragments together with phosphodiester bonds.
11. The overall result is that there are now 2 DNA molecules, each with 1 strand of the old (parent) DNA and 1 newly synthesised DNA strand. Hence the replication of DNA is semi-conservative. 11 max [6]

(b) Describe phenylketonuria (PKU) as an example of gene mutation.

- Autosomal recessive disease
- Phenylalanine →Tyrosine;

- The enzyme responsible for this conversion gets mutated.
- Phenylalanine accumulates. Then,
- Phenylalanine → Phenylpyruvic acid → Accumulates in brain → Mental retardation
- Phenylpyruvic acid also gets excreted through urine since kidneys poorly reabsorb it. [6]

Ref. #9ScB. 9190/2 N2013

8. Table 1 shows the diploid chromosome number in different organisms.

Table 1

Name of organism	Diploid chromosome number 12 n
yeast	34
broad bean	12
fruit fly	8
domestic cat	72
human	46

(a) State the number of chromosomes present in the following:

(i) domestic cat sperm cell 36

(ii) human white blood cell 23

(iii) broad bean sieve tube cell 0

(iv) fruit fly zygote 4 [4]

(b) Fig 3 shows diagrams of an animal cell at various points in the first nuclear division of meiosis. The diagrams are not in the correct sequence.

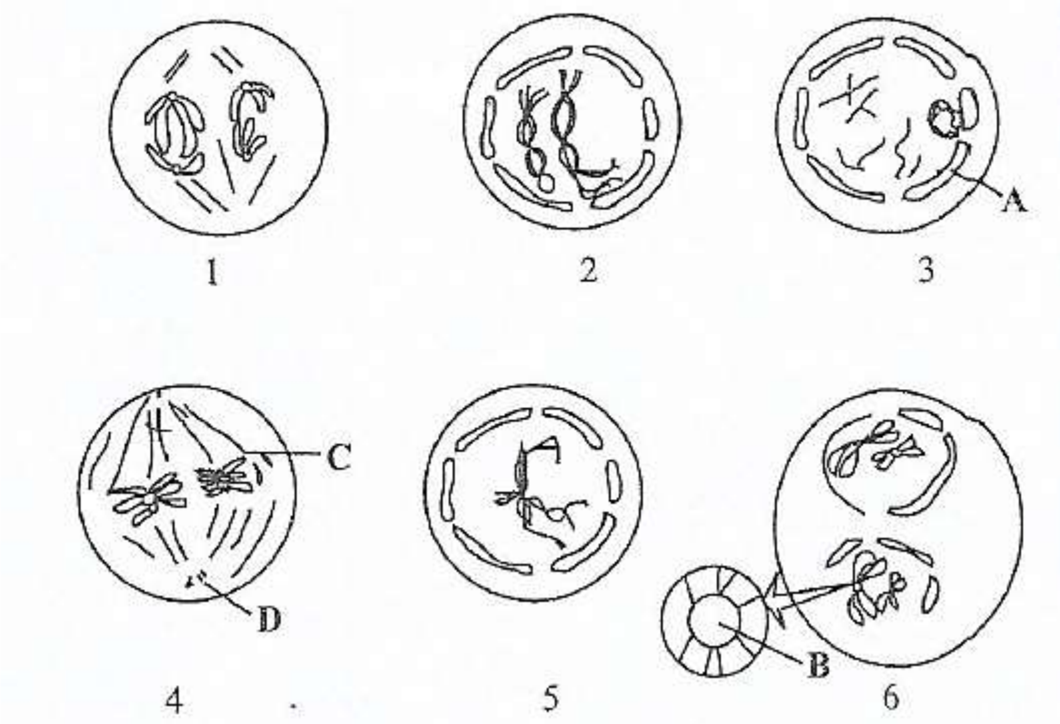


Fig. 3

(i) Use the numbers to show the correct meiotic sequence.

3, 5, 2, 4, 1, 6 [1]

(ii) Identify the structures labelled **A** to **D**.

A- Nuclear envelope/membrane

B- Centromere/kinetochore

C- Spindle fibres

D- centriole(s) [4]

Ref. #3ScA. 9190/2 N2011

9. Mitosis and meiosis are two forms of nuclear division which occur in organisms.

(a) State two similarities between the prophase stage of mitosis and prophase I of meiosis.

Chromosomes shorten and thicken;

Spindle fibres form;

Nuclei disappear. 3 max [2]

(b) Fig 4.1 shows a summary of the stages in the process of meiosis.

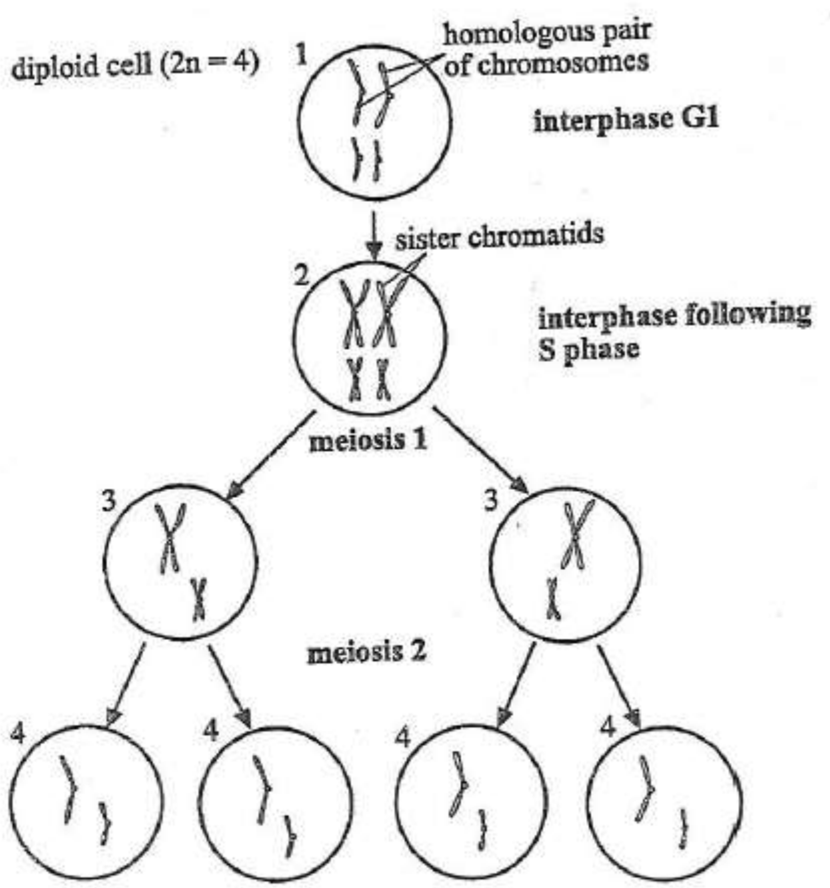


Fig. 4.1

The cell numbered **1** is diploid and the cells numbered **4** are haploid.

State the meaning of the term

(i) *haploid*, Half number of chromosomes/possession of one set of chromosomes (per nucleus/cell) [1]

(ii) *diploid*. Possession of two sets of chromosomes (per nucleus/cell)/full set of chromosomes [1]

(c) Explain the importance of the process of meiosis.

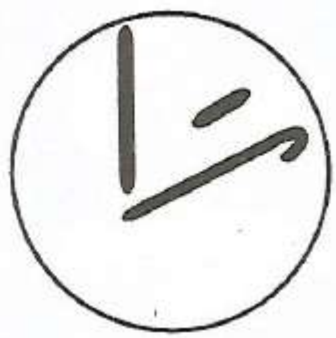
- Increases chances of genetic variation;
- Restores the diploid stage/condition/prevent doubling of chromosomes for successive generation;
- Formation of gametes [4]

Ref. #4ScA. 9190/2 J2015

10. Fig 4.1 shows chromosomes in the nucleus of a diploid cell.

Fig. 4.1

(a) Draw a nucleus of a gamete produced from the cell.



3 chromosomes drawn. 1 from each pair [1]

(b) **Fig 4.2** and **Fig 4.3** show the same diploid nucleus as in **Fig 4.1**. However, the chromosomes have been shaded.

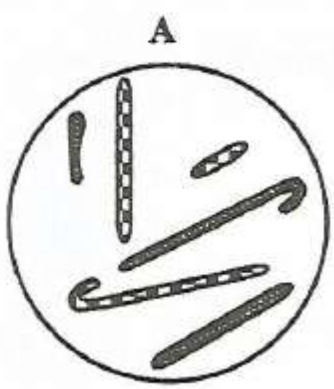


Fig. 4.2

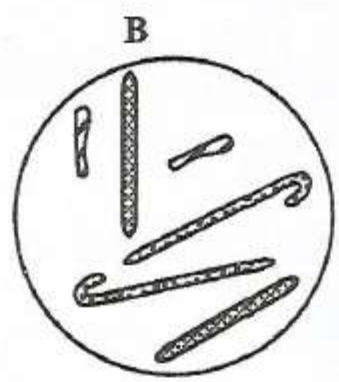


Fig. 4.3

(i) State what is represented by the shading in **Fig 4.2**.

Paternal and maternal chromosomes [1]

(ii) State what is represented by the shading in **Fig 4.3**.

Homologous chromosomes [2]

(c) **Fig 4.4** shows some of the stages that are involved in the development of a human being.

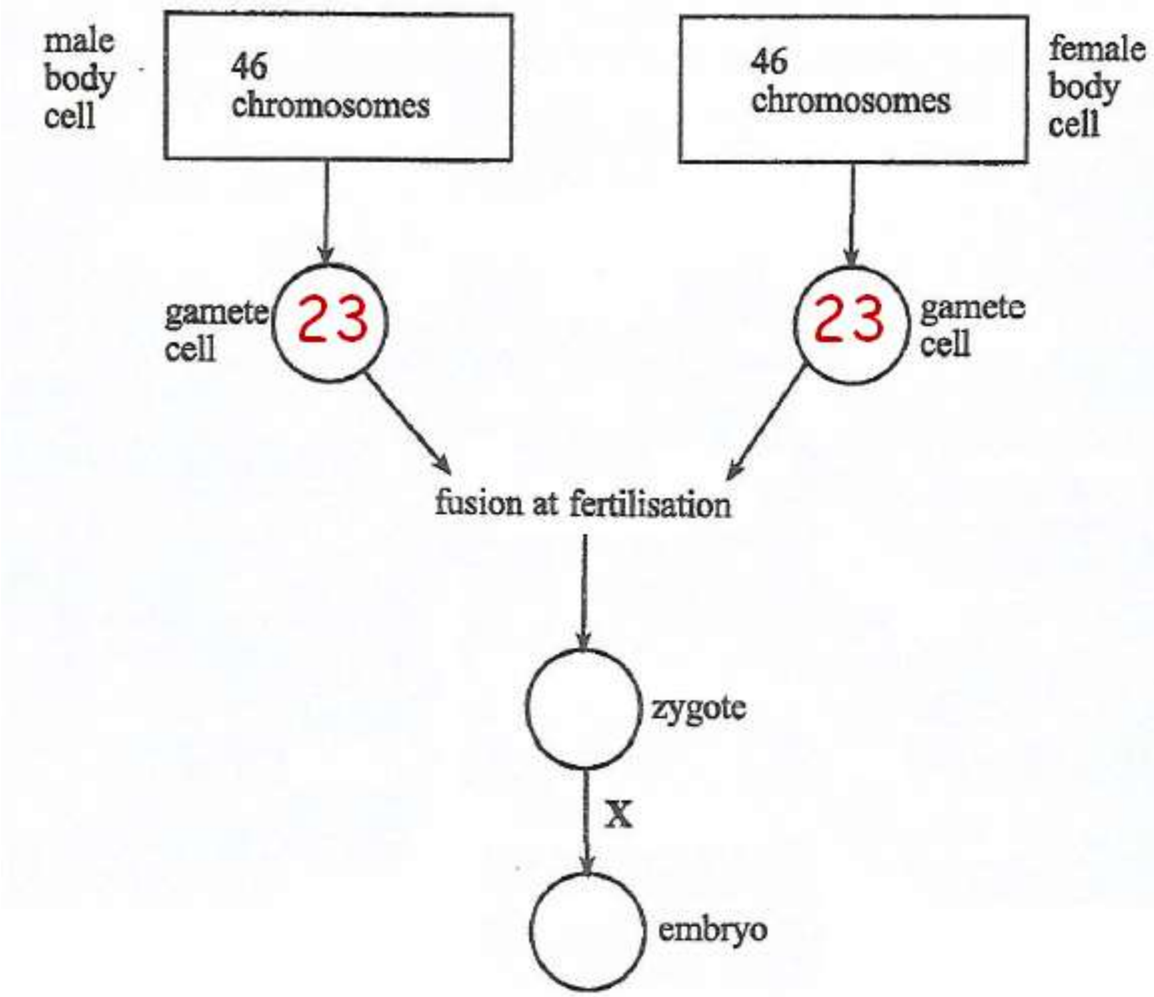


Fig. 4.4

(i) Insert in the circles in **Fig 4.4** the number of chromosomes contained in each gamete cell. [1]

(ii) Identify the type of nuclear division at **X**.

Mitosis

(iii) Explain why meiosis is necessary in the life cycle of sexually reproducing organisms.

Occurs at gamete formation. Each gamete formed is haploid;

Preventing doubling of chromosomes when gametes fuse/maintain genetic stability;

At each generation in species [3]

Ref. #4ScA. 9190/2 J2016

11. (a) Describe and explain how you would prepare a root tip squash to observe the process of mitosis. [8]

(b) Contrast cell division by mitosis in a plant cell and in an animal cell [4]

Ref. #10ScB. 9190/2 N2005 Silver jubilee

12. **(a)** Discuss how meiosis can lead to variation in the offspring. [8]

(b) Outline the factors which increase cancerous growth in an organism. [4]

Ref. #11ScB. 9190/2 N2004

13. **(a)** Discuss the importance of interphase during nuclear division. [4]

(b) Describe the differences between mitosis and meiosis during prophase and telophase stages. [8]

Ref. #12ScB. 9190/2 J2006

14. **Fig 3.1** shows a section through an onion bulb that is starting to grow.

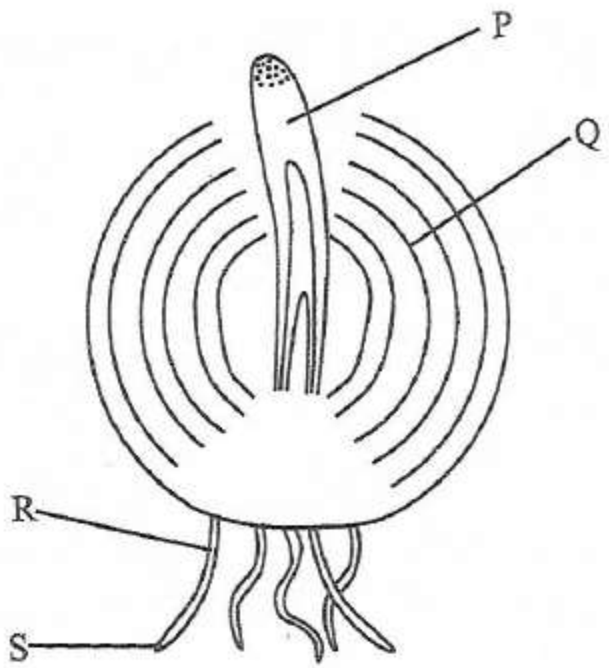


Fig. 3.1

(a) **(i)** Which of the parts labelled P to S of this bulb would be more appropriate to use in preparing a temporary slide to show mitosis.

____ [1]

(ii) Name the stain you would use to make this preparation.

____ [1]

(iii) State the effect this stain would have on the cells.

____

____ [1]

(b) Explain the significance of mitosis to plants.

____

____ [2]

(c) Nma e the phase in the mitotic cell cycle in which the following events would take place:

(i) DNA replication

____ [1]

(ii) Movement of each copy of the chromosomes to the poles of the cell.

____ [1]

Ref. #3ScA 9190/2 N2008

15. **Fig. 5** shows two cells from the same animal in the process of division.

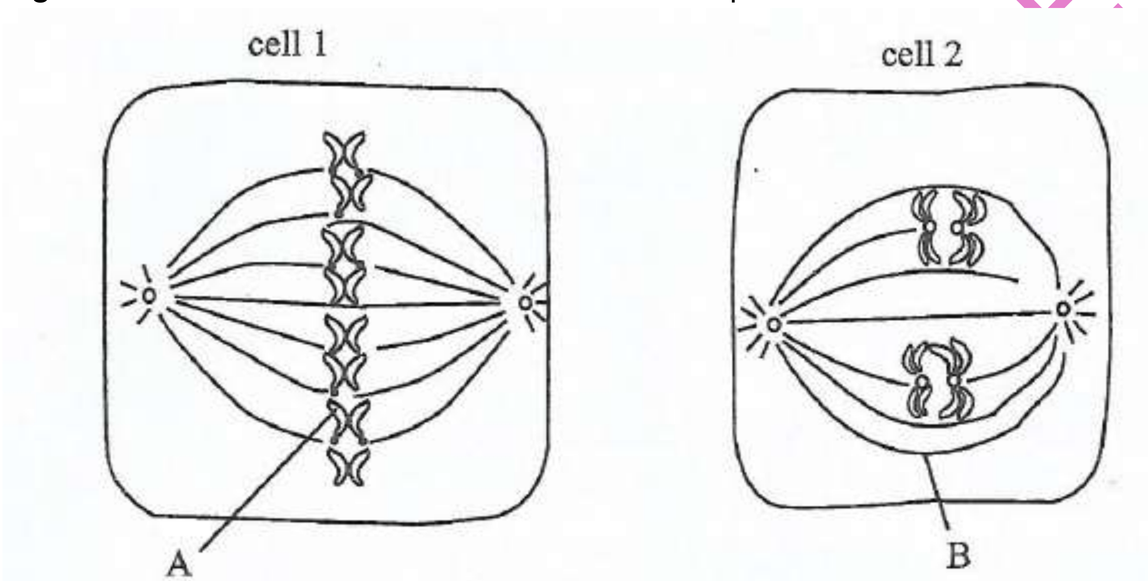


Fig. 6

(a) Name the structures labelled **A** and **B**.

A. Chromatid

B. Spindle fibre [2]

(b) What type and stage of division is shown by **cell 1**?

(i) Type of division: Mitosis [1]

(ii) Stage of division: Metaphase [1]

(c) **(i)** What is the significance of the division in cell 2 to the animal?

Is important in gamete formation

Allows continuation of a species.

Gives rise to variation. [2]

(ii) Suggest one organ in the animal where you expect to find **cell 2**?

In ovary/testis. [1]

Ref. #7ScA. 9190/2 N2009

TOPIC 4 GENETIC CONTROL

1. **(a)** State any three differences between RNA and DNA.

RNA	DNA
1. Single stranded/single polynucleotide chain	Double stranded/double polynucleotide chain
2. Pentose sugar is ribose	Pentose sugar is deoxyribose
3. Base uracil	Base thymine
4. Less stable	More stable (H-bonding)
5. Ratio of adenine: uracil and cytosine: guanine is variable	Ratio of 1:1 for adenine: uracil and cytosine: guanine.
6. Three basic forms: mRNA, tRNA and rRNA	Only one basic form

ANY THREE [3]

(b) Explain how the structure of DNA is adapted to its function.

- Phosphate backbone increase stability/H-bonding increase stability
- Sequence of bases allows DNA to store or carry coded (genetic) information for making proteins/enzymes;
- Very long so that it stores lots of information.
- helical and compact so that lots of information is stored in a small space;
- Complementary base pairing allows the molecule to replicate itself accurately.
- Hydrogen bonds to allow unwinding during replication/transcription [3]

Ref. #2. 6030/2 J2020 NC

2. **Fig 4.1** shows the molecular arrangement of deoxyribonucleic acid (DNA).

(a) Indicate, on **Fig 4.1**, the hydrogen bonds that link the nitrogenous bases. [2]

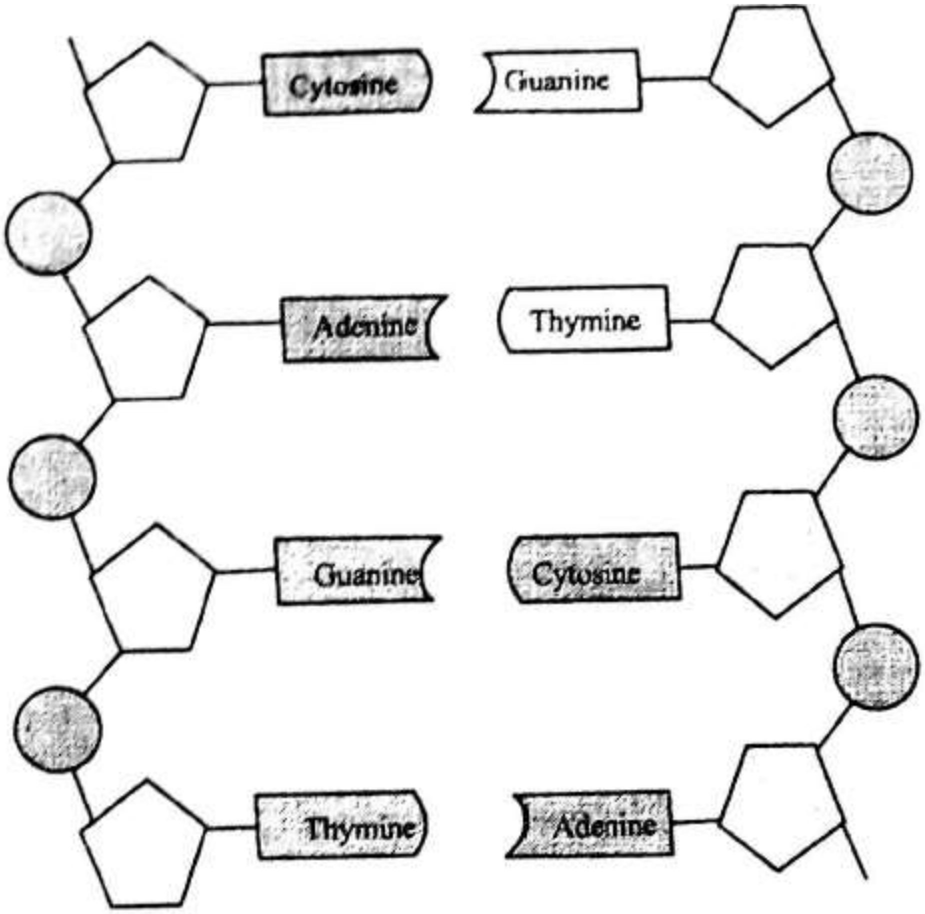


Fig. 4.1

(b) State any two visible features in **Fig 4.1** which make a DNA molecule adapted to carry out its functions.

1 ____

2 ____ [2]

(c) Explain why DNA should be chemically stable.

Every cell must begin with a very accurate copy of its parent's or parents' DNA. So it is important for the DNA to be stable, to resist change, otherwise inaccuracies will appear. [2]

Ref. #4. 6030/2 J2019 NC

3. During protein synthesis, amino acids are carried by tRNA. Each amino acid is first joined to its tRNA by an enzyme. **Fig 4.1** shows the stages in this process.

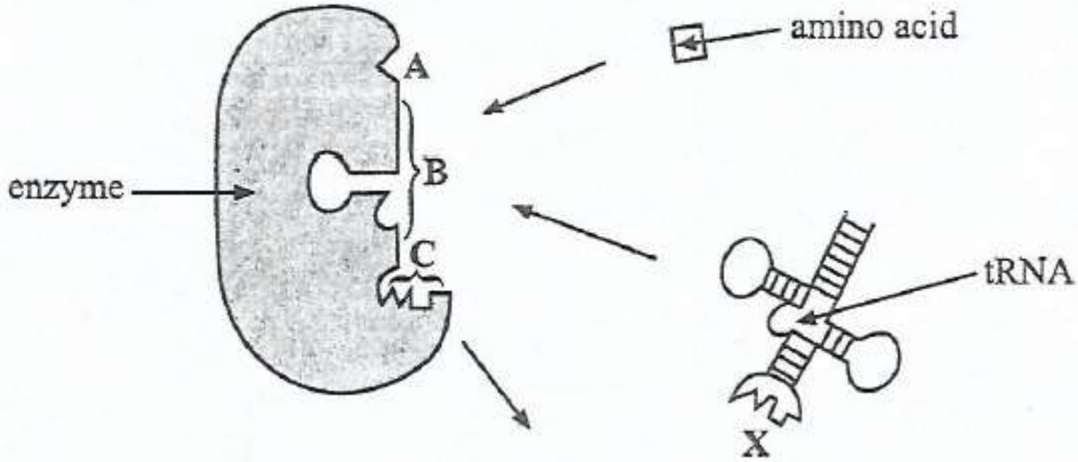


Fig. 4.1

(a) (i) Identify the region **ABC**.

Active site [1]

(ii) The shape of the enzyme molecule partly depends upon different types of bonds.

Name two of these bonds.

1 Peptide bond
2 Hydrogen bonds
3 Disulphide bridges
4 hydrophobic interactions
5Hydrophilic interactions
5 Van der Waals forces ANY TWO [2]

(b) Explain what ensures that the correct amino acid is added during translation.

- Each type of tRNA has a specific linkage;
- with one type of amino acid;
- tRNA joins to mRNA;
- by matching anticodon and codon [3]

Ref. #4 ScA. 9190/2 J2017

4. Fig 4.1 shows a tRNA molecule with part of its structure showing complementary base pairs.

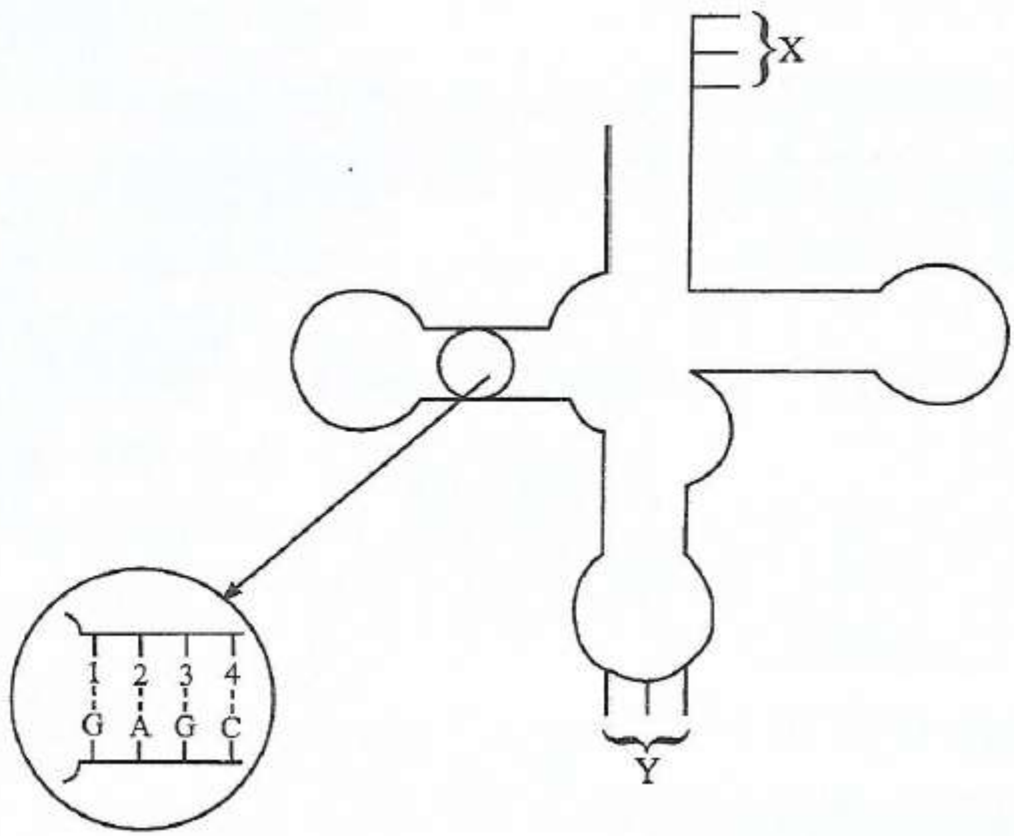


Fig. 4.1

(a) (i) Identify the bases **1, 2, 3** and **4** in **Fig 4.1**.

C U C G [1]

(ii) Name the part **Y**.

Anticodon [1]

(iii) State the role of **X** and **Y** in protein synthesis.

X -site of amino acid attachment

Y -site of attachment to mRNA. [2]

(b) A sample of DNA contains 28% of adenine.
Calculate the amount of cytosine in this DNA sample.

28% A = 28% T

28% x 2 = 56% or 28% + 28% = 56%

100% - 56% = 44%

$\frac{44}{2}$ for C and G + 22%

Therefore Cytosine = 22%.

Amount of cytosine = 22%.

(c) Name a compound containing adenine which is

(i) An energy source in the cell,

Adenosine triphosphate (ATP) [1]

(ii) A hydrogen acceptor.

Nicotinamide adenine dinucleotide (NAD)/NADP/FAD [1]

Ref. #4ScA. 9190/2 N2016

5. Explain how DNA replicates semi-conservatively during interphase.

1. **Helicase** unwinds and separates the double-stranded DNA. This creates a replication fork of two strands running in antiparallel directions.
2. DNA gyrase relieves strain created by the unwinding of DNA by helicase.
3. SSB proteins (single stranded binding proteins) prevents separated strands from re-annealing/ SSB proteins also prevent the single stranded DNA from being digested by nucleases.
4. DNA Primase generates RNA primer to initiate DNA synthesis.
5. DNA polymerase III extends new strand in the 5'→3' direction by joining nucleotides together using energy from hydrolysis (breaking down) of dNTP.
6. DNA polymerases also proofread newly made DNA, replacing any incorrect nucleotides. This ensures that each new DNA double helix is exactly like the original.
7. The leading strand is synthesised continuously towards the replication fork.
8. The lagging strand is synthesised away from the replication fork in short pieces called Okazaki fragments.
9. DNA polymerase I remove RNA primers on lagging strand and replace them with DNA nucleotides.
10. DNA ligase joins Okazaki fragments together with phosphodiester bonds.
11. The overall result is that there are now 2 DNA molecules, each with 1 strand of the old (parent) DNA and 1 newly synthesised DNA strand. Hence the replication of DNA is semi-conservative. 11 max [8]

Ref. #11(b) ScB. 9190/2 N2016

6. (a) Fig 5.1 shows some molecules involved in protein synthesis.

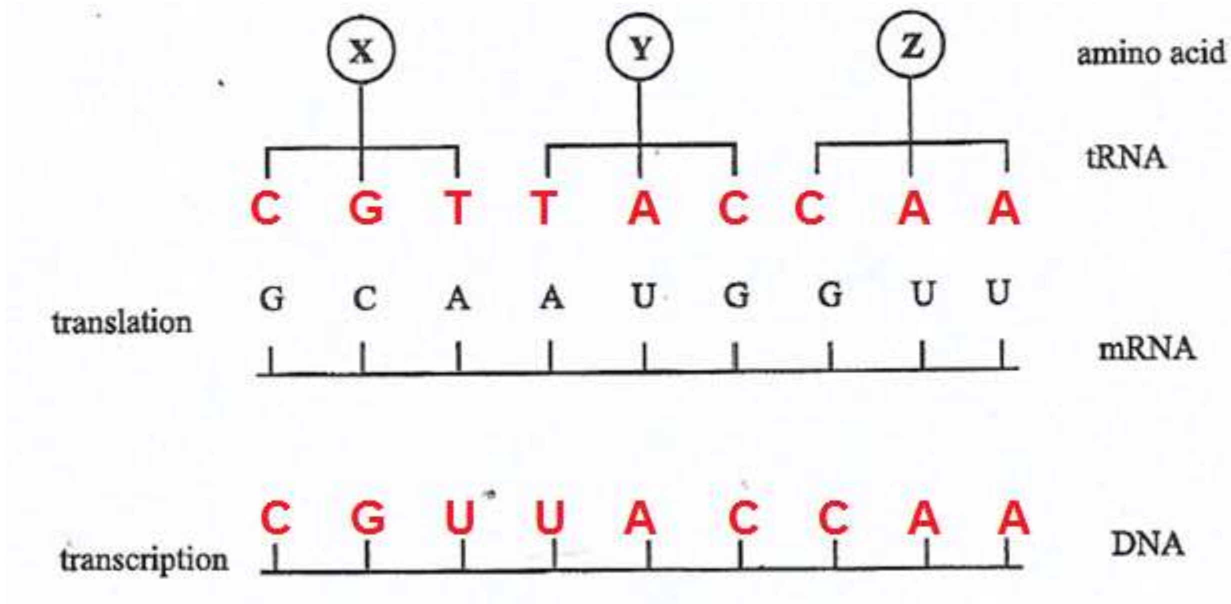


Fig.5.1

(a) **(i)** Name the enzyme involved in transcription.

RNA polymerase [1]

(ii) Name the organelles in which translation occurs.

Ribosomes/Rough endoplasmic reticulum (RER) [1]

(iii) Complete **Fig 5.1** to show

1 the bases on DNA strand,

2 the bases forming the anticodons on the tRNA molecules. [2]

(b) Phenylketonuria is a genetic disease caused by mutation of a gene which codes for the enzyme phenylalanine hydroxylase (PAH). This results in high concentrations of phenylalanine which may be harmful to nerve tissue. Table 5.1 shows part of the DNA base sequence coding for PAH and also a mutation of this sequence which leads to the production of non-functional PAH.

Table 5.1

DNA base sequence coding for PAH	C	A	G	T	T	C	G	C	T	A	C	G
DNA base coding for non-functional PAH	C	A	G	T	T	C	C	C	T	A	C	G

(i) Show, by ticking on **Table 5.1**, the position of the mutation. [1]

DNA base sequence coding for PAH	C	A	G	T	T	C	G ✓	C	T	A	C	G
DNA base coding for non-functional PAH	C	A	G	T	T	C	C ✓	C	T	A	C	G

(ii) State the maximum number of amino acids for which this base sequence could code.

Four/4 [1]

(c) Explain why this mutation may lead to the formation of non-functional PAH.

Change in amino acid sequence;
Change in hydrogen/ionic/disulphide bonds;
Alters tertiary structure/active structure of enzyme
Substrate not complementary/cannot bind to active site. [3]

Ref. #5ScA. 9190/2 N2015

7. Compare and contrast the structure of DNA and RNA.

RNA	DNA
Single polynucleotide chain/strand	Double polynucleotide chain/chain
Very long strands	Relatively short strands
May have a single or double helix	Always double helix
Pentose sugar is ribose	Pentose sugar is deoxyribose
Organic base present is adenine, guanine, cytosine and uracil	Organic base present is adenine, guanine, cytosine and thymine
Ratio of adenine : uracil and cytosine : guanine is variable.	Ratio of 1;1 for adenine : uracil and cytosine : guanine.
Three basic forms; mRNA, tRNA and rRNA	Only one basic form

7 max [6]

Ref. #11(b)ScB. 9190/2 N2011

8. **(a)** Describe the structure of RNA.

- Consist of nucleotide monomers
- Linked by phosphodiester bonds
- Each nucleotide monomer consist of pentose sugar + nitrogenous base + phosphoric acid/phosphate group
- Phosphodiester bond forms between phosphoric acid group of one nucleotide and hydroxyl group of the 3' carbon atom on pentose ring;
- Nitrogenous base may either be pyrimidine (single ring) or purine (double ring);
- Nitrogenous bases in RNA are guanine, adenine, cytosine and uracil;
- RNA may fold back on themselves to form hairpin loops/fold of complementary base pairs;
- RNA molecules are single stranded;

- There are 3 types of RNA namely mRNA, tRNA and rRNA.
- RNA molecules are small(rarely exceeding 10 000 base pairs) 10 max [8]

(b) Outline how DNA replicates semi-conservatively.

1. **Helicase** unwinds and separates the double-stranded DNA. This creates a replication fork of two strands running in antiparallel directions.
2. DNA gyrase relieves strain created by the unwinding of DNA by helicase.
3. SSB proteins (single stranded binding proteins) prevents separated strands from re-annealing/ SSB proteins also prevent the single stranded DNA from being digested by nucleases.
4. DNA Primase generates RNA primer to initiate DNA synthesis.
5. DNA polymerase III extends new strand in the 5'→3' direction by joining nucleotides together using energy from hydrolysis (breaking down) of dNTP.
6. DNA polymerases also proofread newly made DNA, replacing any incorrect nucleotides. This ensures that each new DNA double helix is exactly like the original.
7. The leading strand is synthesised continuously towards the replication fork.
8. The lagging strand is synthesised away from the replication fork in short pieces called Okazaki fragments.
9. DNA polymerase I removes RNA primers on lagging strand and replace them with DNA nucleotides.
10. DNA ligase joins Okazaki fragments together with phosphodiester bonds.
11. The overall result is that there are now 2 DNA molecules, each with 1 strand of the old (parent) DNA and 1 newly synthesised DNA strand. Hence the replication of DNA is semi-conservative. 11 max [8]

Ref. #9ScB. 9190/2 J2015

9. **(a) (i)** List two structural differences between DNA and RNA.

	DNA	RNA
1	______________________	______________________
	______________________	______________________
2	______________________	______________________
	______________________	______________________ [2]

(ii) In the space below draw a labelled diagrammatic representation of a DNA nucleotide.

[2]

(b) Fig 3.1 shows some of the results and interpretation of Meselson and Stahl's experiment on DNA replication.

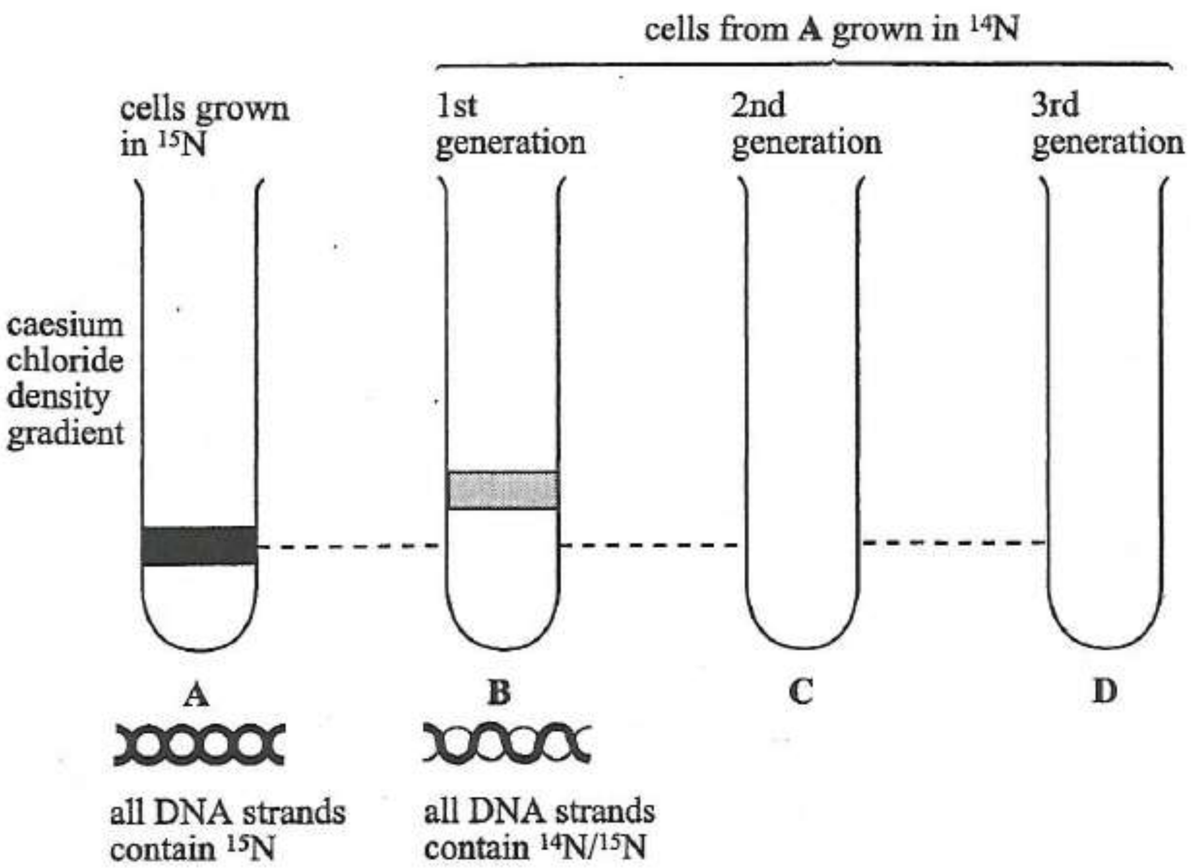


Fig. 3.1

(i) Name the process of DNA replication shown by Fig 3.1.

(ii) Draw on the diagram bands in centrifuge tubes C and D to reflect the amounts of the various types of DNA molecules. [3]

(iii) Describe the nitrogen composition of DNA in centrifuge tubes **C** and **D**.

C ____

____

____

D ____

____

____ [3]

Ref. #3ScA. 9190/2 N2005 Silver jubilee

10. Deoxyribonucleic acid (DNA) is a polynucleotide consisting of three chemical groupings.

(a) Name the **three** chemical groupings within each nucleotide.

- Deoxyribose sugar;
- Phosphate/phosphoric acid;
- Nitrogenous/organic base [3]

(b) State the differences between the four types of nucleotide in DNA.

These derive from the four different organic bases;

Adenine and guanine which are purines, and cytosine and thymine which are pyrimidines. [2]

(c) Describe the semi-conservative replication of DNA.

DNA double helix unwinds/unzip by breaking of hydrogen bonds;

Pairing of free nucleotides with their complementary organic bases on the two separate DNA strands.

Forming two new double helices ; each with one old and one new strand [2]

(d) Explain the importance of the production of two exact copies of the original DNA molecule during DNA replication.

To maintain genetic stability/Daughter cells produced by mitosis (during growth and repair) will be identical to parent cells;

So that they function like the cells they complement of replace. [2]

Ref. #3ScA 9190/2 N2009

TOPIC 5 GENE TECHNOLOGY

1. **(a) (i)** Define genetic screening.
   Analysis of person's DNA to check for abnormalities/genetic disease [1]
   (ii) State the two methods used in prenatal genetic screening.
   1 Chorionic villi sampling
   2 Amniocentesis. [2]

(b) Describe the advantages of genetic screening.

- Reduce the incidence of genetic diseases;
- Change of life style if predisposed;
- Reduce suffering of victim/family;
- Reduce economic cost to a country. [4]

Ref. #6. 6030/2 J2020 NC

2. **(a)** Genetically modified mammalian cells from the ovary or kidney can be used to produce a glycoprotein, human factor VIII, required for blood clotting.

State **two** advantages of treating haemophiliacs with genetically engineered human factor VIII.

1 There is greater rate of production;
2 No contamination or infection e.g. with HIV;
3 No side effects ;
4 Fewer ethical objections [2]

(b) Insulin is injected into people who are diabetic. Explain why it is injected and not taken orally.

- Hormone is protein;
- So would be digested (in alimentary canal);
- Travels quickly to target organs (in blood) 3max [2]

(c) State any two non-biological methods, other than the use of plasmids, of introducing genes into host cells.

1. Liposome transfer;
2. Electroporation;
3. Microinjection/micropipette;
4. A DNA gun fires tungsten or gold particles coated with DNA/ ballistic impregnation. 4 max [2]

(d) Explain why mammalian cells are used as host cells in recombinant DNA technology during production of human factor VIII.

- Human genes contain exons and introns;
- Mammalian cells have enzymes to remove introns/enzymes for methylation and splicing of mRNA;
- Mammalian cells have enzymes/golgi apparatus for post-translational modification;
- Presence of human regulator genes. 4 max [3]

Ref. #4ScA. 9190/2 N2014

3. (a) Describe the formation of recombinant DNA.

- DNA formed in vitro by linking DNA fragments from different organisms;
- Identify required gene/synthesize it artificially/isolate it;
- Use reverse transcriptase to manufacture cDNA from mRNA;
- Use restriction endonuclease enzyme to cut the DNA at specific points into restriction fragments;
- Find required gene through gel electrophoresis;
- Cut plasmid DNA using same restriction enzyme;
- Insert isolated gene into a vector such as plasmid DNA and phage DNA;
- Use DNA ligase to attach to cDNA; [8]

(b) Discuss the potential hazards of gene technology with reference to crop plants.

- Modified crop plants may become agriculture weeds;
- Introduced genes may be transferred by pollen to wild relatives;
- Hybrid offspring may become more invasive;
- Introduced genes may be transferred by pollen to unmodified plants; growing on farm with organic certification;
- The modified plants may be a direct hazard to humans/domestic animals ; by being toxic/producing allergies;
- The herbicide that can now be used on crops will leave toxic residue;
- Danger of losing traditional varieties (with genes) for particular localities [8]

Ref. #10ScB. 9190/2 J2016

4. Biologists have learned much about cell structure and function by studying cells and fragments of tissue removed from multicellular organisms. He La cells were established in 1951 from a young black woman called Hentietta Lacks who had cervical cancer. He La cells served as culture systems for numerous viruses, including polio virus. Later these cells were made to synthesise mouse proteins. There are several ways of including this process, one of which involves the use of bacterial plasmids as vectors and restriction enzymes as is ythe case in the synthesis of human insulin by bacteria.

(a) (i) What is the role of restriction enzymes?

____ [1]

(ii) What are bacterial plasmids?

____ [1]

(b) (i) State briefly why restriction enzymes are used in genetic engineering.

____ [2]

(ii) What other types of enzymes are used in this process?

____ [1]

(c) Explain why it is possible that a gene in the mouse can be used to produce the same protein in the He La cells.

____ [4]

Ref. #3ScA. 9190/2 N2004

5. **Fig 4.1** summarises stages involved in genetic engineering.

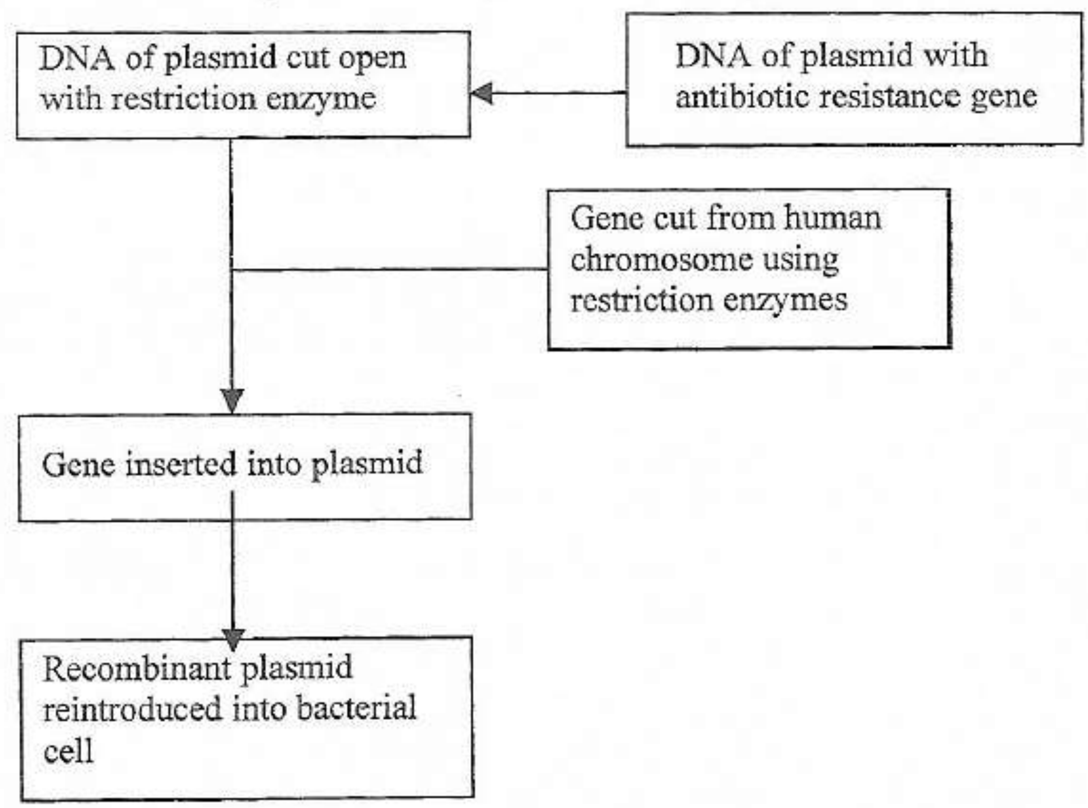


Fig. 4.1

(a) Name the enzyme that is used to insert the human gene into the plasmid.

____ [1]

(b) Describe **one** method that can be used to identify bacterial cells that have succefully taken up the recombinant plasmid.

____

____

____

____ [2]

(c) State the reasons why the same restriction enzyme is used to cut the plasmid and the human gene.

____

____

____ [3]

Ref. #4ScA. 9190/2 N2008

TOPIC 6 INHERITED CHANGE AND EVOLUTION

1 **Fig 4.1** shows the inheritance of red-green colour blindness in two families. The condition is sex-linked. The allele for red-green colour blindness is recessive to the allele for normal colour vision.

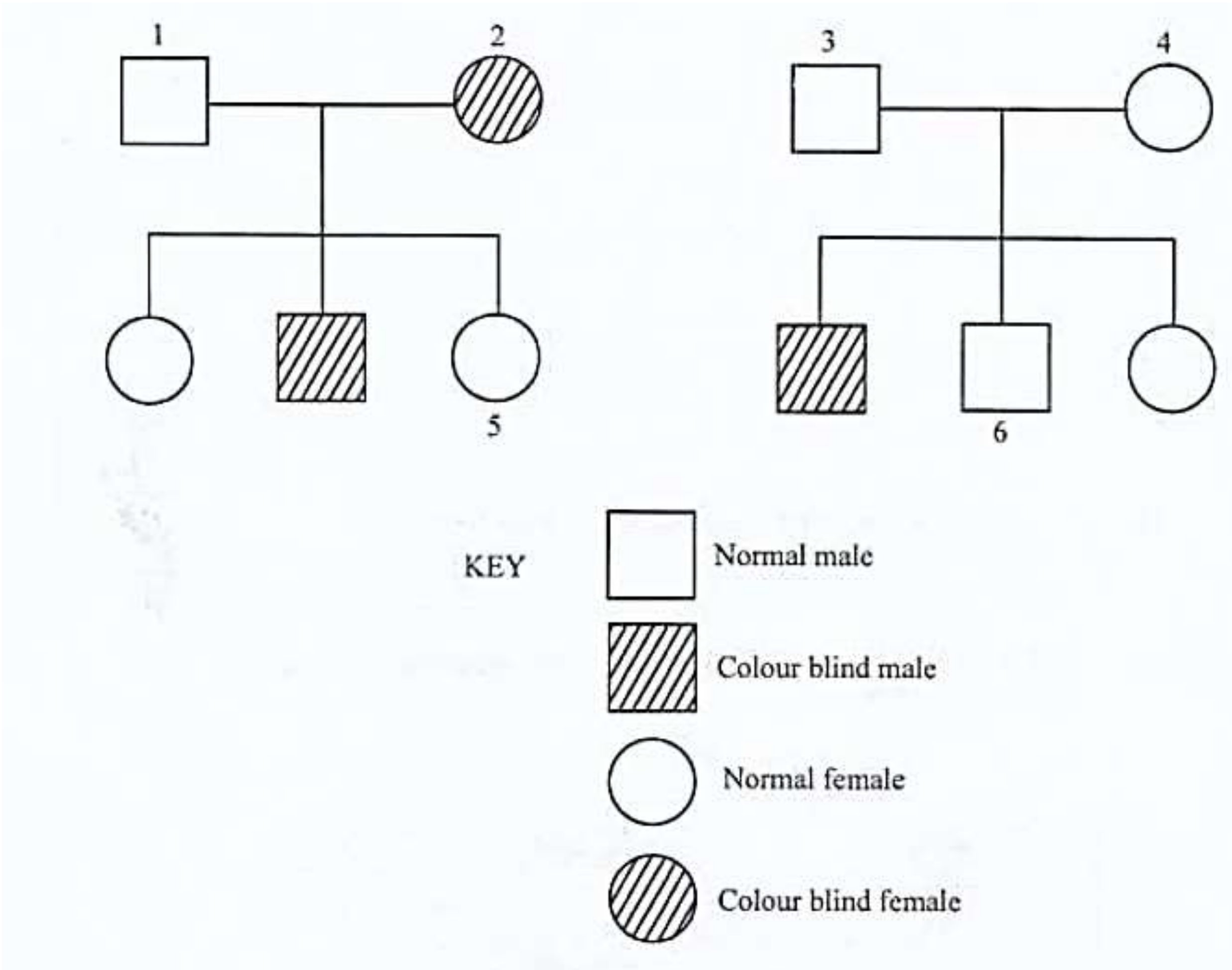


Fig.4.1

(a) Using symbols X^B for normal colour vision and X^b for red-green colour blindness, state the genotypes of

(i) Individual **2.** ____________________

(ii) Individual **4.** ____________________ [2]

(b) Using a genetic diagram, determine the possible genotypes and phenotypes of children that may be produced by individuals **5** and **6**. [4]

Ref. #4ScA. 9190/2 J2018

2 (a) Name the process which produces mRNA from DNA.

____________________ [1]

(b) Table 5.1 shows some codons on mRNA and the amino acid coded for by each codon.

Table 5.1

Codon	Amino acid
UAC	Tyrosine
AGU	Serine
GAG	Aspartic acid
CAU	Glutamine
CUA	Histidine
UUG	Leucine

State the:
(i) tRNA anticodon for histine. ______________________________
(ii) DNA codon for serine. ______________________________ [2]

(c) Explain why a strand of mRNA with 64 codons produces a protein with only 63 amino acids.
______________________________ [1]

(d) Explain why:
(i) DNA needs to be chemically stable.
____
______________________________ [2]
(ii) mRNA needs to be easily broken down.
____
______________________________ [2]

Ref. #5ScA. 9190/2 J2018

3 One variety of domestic chickens has either black feathers or barred feathers. Barred feathers are black with white bars. The feather colour is controlled by a gene on the X chromosome. The dominant allele B, results in barred feathers while the recessive allele, b, results in feathers that are black. Females are heterogametic while males are homogametic.

(a) A breeder crossed a black female with a barred male. Fourteen offspring were produced:

3 barred males
4 barred females
4 barred males
3 black females

Using a genetic diagram, show this cross.

parental phenotypes	black female		x	barred male	
parental genotypes	X^bY		x	X^BX^b	
gametes	X^b	Y	x	X^B	X^b
offspring genotypes	X^BX^b	X^BY		X^bX^b	X^bY
offspring phenotypes	barred male	barred female		black male	black female; [4]

(b) It is difficult to tell the sex of newly hatched chicks. The breeder wishes to have an easy method by which he could distinguish the male chicks from female chicks.
Draw a genetic diagram to explain how the breeder could do this.

correct cross is a barred female **x** black male;

parental genotypes	X^B	x	X^bX^b;
gametes	X^B Y		all X^b
offspring genotypes	X^BX^b	and	X^bY;
phenotypes	barred males	and	black females;

4 [max 3]

Ref. #5ScA. 9190/2 J2017

4 Describe the role of natural selection in evolution.

1. Individuals in a population have great reproductive potential;
2. Numbers in a population remain roughly constant;
3. Variation in members of a given population;
4. Environmental factors/named factor (biotic or abiotic) act as selection pressure;
5. (cause) many; many fail to survive/die/do not reproduce;
6. Those best adapted survive/ there is survival of the fittest;
7. They reproduce and pass on their advantageous/desirable alleles/.genes;
8. Genetic variation leads to change in phenotypes;
9. There is changes in gene pool/allele frequency;
10. This overtime produces evolutionary change
11. New species arise from existing ones/speciation;
12. Disruptive/stabilising selection/disruptive   12 max [8]

Ref. #11(a) ScB. 9190/2 N2016

5 A pure breeding female fruitfly, *Drosophila melanogaster*, with normal wings and grey body was matted with a pure breeding male fruitfly with vestigial wings and ebony body. All the $\mathbf{F_1}$ offspring had normal wings and grey body. One of the $\mathbf{F_1}$ hybrid male was crossed with a female with vestigial wings and ebony body. The following offspring were produced.

55 Normal wings, grey body
51 Normal wings, ebony body
49 Vestigial wings, grey body
53 Vestigial wings, ebony body

Use the symbols **WW** for normal wing genotype, **GG** for grey body genotype, **ww** for vestigial wings and **gg** for ebony body.

(a) Draw a genetic diagram to explain the results of the second cross. [5]

Parental phenotype:	**Normal wing grey body**	**X**	**Vestigial wing ebony**			
Parental genotype:	**WwGg**	**X**	**wwgg**			
Gametes:	**WG**	**Wg**	**wG**	**wg**	**X**	**wg**
Offspring genotype:	**WwGg**	**Wwgg**	**wwGg**	**wwgg**		
Offspring phenotype:	**Normal wing. grey body.**	**Normal wing. ebony body.**	**Vestigial wing. grey body.**	**Vestigial wing. ebony body**		
Phenotypic ratio:	**1 :**	**1 :**	**1 :**	**1**		

max [5]

(b) Use the formula $\chi^2 = \sum \frac{(O-E)^2}{E}$

$\sum$ = 'Sum of'
O = Observed value
E = Expected value

to test for the F_2 phenotypic ratio obtained agrees with the expected 1:1:1:1 ratio.

χ^2 distribution

	probability (p)								
degrees of freedom	0.99	0.95	0.80	0.70	0.50	0.30	0.20	0.005	0.001
1	0.0016	0.004	0.064	0.148	0.455	1.074	1.6421	3.841	6.635
2	0.103	0.446	0.713	0.020	1.386	2.408	3.219	5.991	9.210
3	0.115	0.352	1.005	1.424	2.366	3.665	4.642	7.815	11.341

Test

Total number of fruit fly = 55 + 51 + 49 + 53 = 208

Total ratio = 1 + 1 + 1 + 1 = 4

Expected number of fruit fly in each case is the same = $\frac{1}{4}$ x 208 = 52

Phenotype	Observed (O)	Expected (E)	O – E	$(O - E)^2$	$\frac{(O - E)^2}{E}$
Normal wings, grey body	55	52	3	9	0.173
Normal wings, ebony body	51	52	-1	1	0.019
Vestigial wings, grey body	49	52	-3	9	0.173
Vestigial wings, ebony body	53	52	1	1	0.019
				χ^2 =>	**0.384**

Degrees of freedom (df) = 4 – 1 = 3

X2 value falls between probability levels of p = 0.95 and p =0.80./ Tabulated result is greater than calculated value.

Conclusion

There is a greater than 80% probability that the deviation from the expected occurred by chance.

Therefore data agree with our 1:1:1:1 ratio/Accept null hypothesis (that F2 phenotypic ratio agrees with the expected 1:1:1:1 ratio). [4]

Ref. #5ScA. 9190/2 N2014

6 A tomato plant with the phenotype purple stems and cut leaves, but of unknown genotype, was crossed with a tomato of double recessive genotype for green stems and uncut leaves. The cross produced the following results:

74 plants with purple stems and cut leaves
80 plants with green stems and cut leaves.

(a) Draw a genetic diagram to explain this cross.
Use the letter **G** for purple stems and **g** for green stems, **H** for cut leaves and **h** for uncut leaves,

Parental phenotype:	Purple stem, cut leaves	X	Green Stem, uncut leaves
Parental genotype:	GgHH	X	gghh
Meiosis:			
Gametes:	GH gH	X	gh
Fertilisation:			
F1 genotype:	GgHh		ggHh
F1 phenotype:	Purple stem, cut leaves.		Green stem, cut leaves
Phenotypic ratio:	1 : 1		**[4]**

(b) Explain how a plant breeder could produce a pure breeding tomato plant with purple stems and cut leaves from the plants described in the passage above.

- In breeding of purple cut plants from F1;
- Test crossing plants with purple stem, cut leaves with pure breeding green uncut plants to identify pure breeding purple, cut plants;
- In breeding of pure breeding purple stem cut leaves plants for several generations. [3]

Ref. #4ScA. 9190/2 N2013

7. Two pure breeding strains of snap-dragon, a garden plant, were obtained. One strain had red flowers and the other had white flowers. The two strains were crossed yielding F_1 plants with pink flowers. The plants were then interbred to produce F_2 plants with the following petal colours.

Red: 62
Pink: 131
White: 67

The following hypothesis was proposed:

Flower colour is controlled by a single gene with two codominant alleles.

(a) Complete the genetic diagram, using the symbols given to represent the alleles, to explain this cross.

C^r = red; C^W = White

Parental phenotypes	Red flowers	x	White flowers		
Parental genotypes	$C^r C^r$		$C^W C^W$		
Gametes	C^r		C^W		[1]
F_1 genotypes	All $C^r C^W$				
F_1 phenotypes	All pink				
Gametes	C^r	C^W	C^r	C^W	[1]
F_2 genotypes	$C^r C^r$	$C^r C^W$	$C^r C^W$	$C^W C^W$	
F_2 phenotypes	red	pink	pink	White	
Expected F_2 phenotype ratio	1 : 2 : 1				[1]

(b) A chi-squared (χ^2) test is carried out on the experimental data to determine whether the hypothesis is supported.

(i) Complete **Table 2** by calculating the expected numbers.

Table 2

F2 phenotype	Observed numbers	Expected numbers
Red	62	65
Pink	131	130
White	67	65
Total	260	260

[2]

Total ratio = 1+2+1 = 4

Expected Pink = 2/4 x 260 = 0.5 x 260 = 130

Expected white = 1/4 x 260 = 65

The χ^2 statistic is calculated in the following way

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

where $\sum$ = 'Sum of'
O = Observed value
E = Expected value

(ii) Calculate the χ^2 value for the above data. Show your working.

$$\frac{(62-65)^2}{65} + \frac{(131-130)^2}{130} + \frac{(67-65)^2}{65}$$

$$= \frac{(-3)^2}{65} + \frac{1^2}{130} + \frac{2^2}{65}$$

$$= \frac{9}{65} + \frac{1}{130} + \frac{4}{65}$$ **ACCEPT** decimals 0.138 +0.008 + 0.062

$$= \frac{27}{130} = 0.208$$

[2]

(iii) The critical value of χ^2 for this type of investigation with two degrees of freedom is 5.991.

Explain whether your answer to (b)(ii) supports the hypothesis.

Supports the (null) hypothesis

Reason: The calculated χ^2 value (0.208) is less than the critical value (5.991).

(Any deviations are due to chance). [1]

Ref. #6ScA. 9190/2 N2011

8. In fruit flies, *Drosophila melanogaster*, the allele for grey body colour is dominant to lack of body colour, while the allele for normal wing shape is dominant to bent wing shape. A cross was made between a fly heterozygous for both grey body colour and normal wing shape and a fly with black body and bent wing shape. The numbers and phenotypes of the offspring were as follows:

Grey body and normal wing shape	83
Black body and normal wing shape	85
Grey body and bent wing shape	77
Black body and bent wing shape	75

(i) Draw a genetic diagram using the following symbols to represent the alleles:

G = Grey body colour **g = Black body colour**

N = Normal wing shape **n = Bent wing shape.**

Parental phenotypes:	Grey body, normal wing		X		Black body, bent wing		
Parental genotypes:	GgNn		X		ggnn		
Meiosis:							
Gametes:	GN Gn gN gn		X		gn		
Offspring genotypes:	GgNn		Ggnn		ggNn		ggnn
Offspring phenptypes:	Grey body normal wing.		Grey body bent wing.		Black body normal wing		Black body bent wing.
Phenotypic ratio:	1	:	1	:	1	:	1

[5]

(ii) A χ^2 test was then carried out to find out if the results in **(i)** showed that independent assortment had taken place.

Complete **Table 6.1**.

Table 6.1

offspring	observed numbers	expected numbers
grey body, normal wing shape	83	80
black body, normal wing shape	85	80
grey body, bent wing shape	77	80
black body, bent wing shape	75	80
Total	320	320

[1]

Total ratio = 1+1+1+1=4

Each expected number = $\frac{1}{4}$ x 320 = 80

(iii) Calculate the χ^2 value for the above data.

$$\left[\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected} \right]$$

where $\sum$ is 'Sum of'

$$\frac{(83-80)^2}{80} + \frac{(85-80)^2}{80} + \frac{(77-80)^2}{80} + \frac{(75-80)^2}{80}$$

$$= \frac{3^2}{80} + \frac{5^2}{80} + \frac{(-3)^2}{80} + \frac{(-5)^2}{80}$$

$$= \frac{9}{80} + \frac{25}{80} + \frac{9}{80} + \frac{25}{80}$$ /ACCEPT Decimal fractions 0.1125 + 3125 + 0.1125 + 3125

$$= \frac{68}{80}$$

$$= 0.85$$

χ^2 = 0.85 [2]

(iv) The critical value of χ^2 for this type of investigation with three degrees of freedom is 7.82.

Explain whether the observed results were significantly different from the expected results.

The calculated χ^2 value (0.85) is less than 7.82

Therefore, the observed results are not significantly different from the expected.(Any variation is due to chance) [2]

Ref. #6ScA. 9190/2 J2016

9. (a) Humans sometimes grow up with bones that are brittle and break easily. This condition is passed on from parents to their children. **Fig 5.1** shows a family tree with the brittle bone trait.

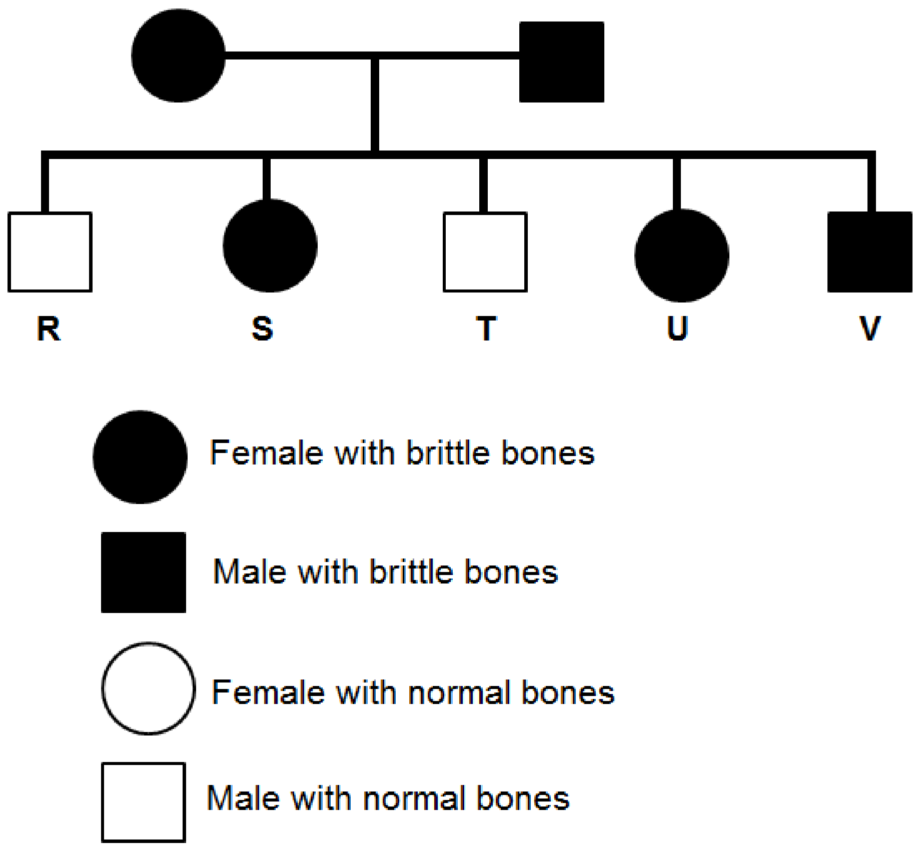


Fig. 5.1

(i) Sate the dominant condition in **Fig 5.1**.

Brittle bones [1]

(ii) The allele for brittle bones is **B** and for normal bones is **b**.
Suggest the possible genotypes for children **S** and **T**.

S X^BX^B or X^BX^b

T X^bY [2]

(b) In fruit flies Drosophila the gene for long wing length and for eye colour are sex-linked. Normal wing and red eye are dominant to miniature wing and white eye.

(i) Define the term *sex-linked*.

When gene is on part of the X chromosome and it is passed on to the next generation. [1]

(ii) A cross between a miniature wing, red-eyed male and homozygous normal wing, white-eyed female was carried out.

Using symbols **N** for normal wing, **R** for red eye, **r** for white eye and **n** for miniature wing, draw a genetic diagram to show the genotypes and phenotypes of the F_1 generation in the space below. [5]

Parental phenotype: Male: miniature wing red-eyed X Female: normal wing white-eyed

Parental genotype: $X^{nR}\ Y$ x $X^{Nr}\ X^{Nr}$

Meiosis:

Gametes: X^{nR}, Y x X^{Nr}, X^{Nr}

Fertilisation:

F1 genotype: $X^{nR}X^{Nr}$ $X^{Nr}Y$ $X^{nR}X^{Nr}$ $X^{Nr}Y$

F1 phenotype: 2 Normal wing red-eyed female : 2 Normal wing white-eyed male

Phenotypic Ratio: 1 : 1

Ref. #5ScA. 9190/2 J2015

10. The inheritance of colour and length of feathers in a certain species of birds is determined by two different genes. Presence of allele **B** results in blue feathers while presence of the recessive allele **b** results in appearance of white feathers. Presence of the allele **C** results in short feathers while presence of the recessive allele **c** results in development of long feathers.

(a) (i) State the possible genotypes of a bird with white and short feathers.

____ [2]

(ii) Describe how you would determine the genotype of a bird with white and short feathers.

____

____

____

____ [4]

(b) A cross between two heterozygous birds with blue and short feathers produces offspring with the following phenotype ratios;

9 blue, short feathers: **3** blue, long feathers: **3** white, short feathers: **1** white, long feathers.

What conclusion can be drawn from the results of the above cross?

____

____

____

____ [2]

Ref. #4ScA. 9190/2 N2005 Silver jubilee

11. **(a)** Explain, with examples, how the environment may affect the phenotype. [6]

(b) Discuss, with examples, the effect of environmental factors acting as forces of natural selection. [6]

Ref. #11ScB. 9190/2 N2005 Silver jubilee

12. The blood group in humans is controlled by an autosomal gene. The condition is known as multiple allele. The alleles are I^A, I^B and I^O.

(a) **(i)** Describe the term *multiple allele*.

____

____ [1]

(ii) Give all the possible genotypes of the human blood groups.

____

____

____

____ [3]

(b) **(i)** Determine, using genetic symbols, the possible blood groups of children where the father is homozygous with respect to **A** allele and the mother is heterozygous blood group **B**.

[3]

(ii) If these parents have identical twins what is the probability that both twins will have blood group A? Explain.

____
____
____
____ [2]

Ref. #4ScA. 9190/2 N2004

13. Red green colour blindness is a sex linked recessive trait in humans. A colour blind man marries a woman with normal vision, whose father is colour blind. **N** is the allele for normal vision, and **n** is the allele for colour blindness.

(a) By means of appropriate symbols, state the genotype of the woman. Explain your answer.

Woman's genotype ____ [1]
Explanation ____
____
____ [2]

(b) **(i)** In the space provided below, construct a genetic diagram to show the genotypes and phenotypic proportions produced by this cross. [4]
(ii) Calculate the chances that the first child from this marriage will be a colour blind boy. [2]
Ref. #7ScA. 9190/2 J2006

14. **(a)** Describe the effects of environmental factors on natural selection. [6]
(b) Discuss the validity of the following statement:
"Artificial selection of crop plants and farm animals has tended to reduce variety within their populations". [6]
Ref. #11ScB. 9190/2 N2008

15. In short-horn cattle the alleles for red and white coat colour are codominant. A cross between red bulls and white cows produces calves with red and white hairs in their coats. These are called "roan".

(a) (i) Define the term *codominance*.

Codominance is whereby two alleles fail to dominate each other in heterozygosis and their effects are expressed together in the offspring. [2]

(ii) Choose suitable symbols to denote the alleles for coat colour in the short-horn cattle.

Red allele - R. White allele - W. [1]

(b) Draw a genetic diagram to show the genotypes and phenotypes of all the cattle and calves in a cross between red bulls and white cows in the space below. [3]

Red allele – R
White allele – W

Parental phenotypes:	Red bull	x	White cow	
Parental genotypes:	RR		WW	
Meiosis:				
Gametes:	R	R	W	W
Random fertilisation:				
F1 genotype:	RW	RW	RW	RW
F1 phenotype:	All roan			

(c) If roan bulls and cows are mated together, what genotypes and phenotypes would you expect in the calves? Use a genetic diagram to illustrate your answer. [4]

Red allele – R.
White allele – W.

Parental phenotypes:	Roan bull	x	Roan cow	
Parental genotypes:	RW		RW	
Meiosis:				
Gametes:	R	W	R	W
Random fertilisation:				
F2 genotype:	RR	RW	RW	WW
F2 phenotype:	Red	Roan	Roan	White
Genotypic ratio:	1 : 2 : 1			
Genotypic ratio:	1 RR : 2 RW : 1 WW			

Ref. #4ScA 9190/2 N2009

TOPIC 7 ENERGETICS

1. **(a)** State the principle of *limiting factors*.

   When a chemical process is affected by more than one factor; the rate of the process is limited by that factor which is nearest its minimum. [2]

   (b) State and explain any two limiting factors of photosynthesis.

   - **Factor 1**- Temperature
     Explanation- Very low temperatures inactivate enzyme; Temperature below optimum increases K.E. of enzyme-substrate complexes; Temperature above optimum denatures enzyme.
   - **Factor 2**- Carbon dioxide concentration
     Explanation- Increase in carbon dioxide concentration increases substrate concentration for enzymes/ more enzyme-substrate complex formed/ increased fixation formation of GP/ REF to Vmax & explanation;
   - **Factor 3**- Light intensity
     Explanation- stomatal aperture increase and more gaseous exchange occurs.

   ANY TWO [2]

   Ref. #5. 6030/2 J2020 NC

2. **Fig 5.1** represents a stage in the process of respiration.